

SCOPE OF WORK

Project Name: **Quality Control Application**

Client Name: **Mr. Feras**

1. Project Objective

The objective of this project is to design, develop, and deliver an enterprise-grade Quality Control and Operational Compliance Platform that enables organizations to standardize, execute, monitor, and continuously improve operational routines across multiple locations, teams, and industries.

The platform is intended to replace fragmented, manual, and paper-based quality control processes with a centralized, digital, and enforceable system that ensures:

- Consistent execution of daily, shift-based, and periodic operational routines
- Accurate, real-time visibility into compliance status across locations and assets
- Structured handling of non-conformances through corrective and preventive actions (CAPA)
- Traceable audit trails suitable for internal governance and external regulatory audits
- Reduced operational risk through automation, alerts, and escalation mechanisms
- This project is not limited to checklist digitization. It aims to deliver a foundational Operational Compliance Management System (OCMS) that combines:
 - Routine execution (checklists, inspections, audits)
 - Physical-digital enforcement (QR codes, sensors, location locking)
 - Workflow automation and CAPA lifecycle management
 - Role-based execution for frontline staff, supervisors, managers, and administrators
 - Scalable multi-tenant SaaS architecture suitable for enterprise deployment

The platform is designed to support multi-industry use cases (such as manufacturing, food & beverage, hospitality, healthcare, logistics, facilities management, and other regulated or semi-regulated environments) through configurable templates, workflows, automation rules, and industry-specific structures, while maintaining a single unified core system.

The ultimate objective is to enable organizations to move from reactive compliance to proactive operational discipline, improving quality, safety, accountability, and audit readiness across their operations.

2. Platform Positioning & Core System Philosophy

Platform Positioning

The platform described in this Scope of Work is not a standalone checklist or inspection application. It is a routine-driven Operational Compliance Management System (OCMS) designed to enforce how work is performed, verified, and corrected across distributed operations.

While the platform includes modules such as checklists, audits, CAPA, sensors, scheduling, and reporting, these components are not independent features. They operate as interconnected parts of a single execution and compliance framework.

The system is designed around the following core principles:

- Routine-first execution: Operational compliance is achieved through repeatable daily, shift-based, and periodic routines rather than ad-hoc inspections.
- Execution enforcement, not self-reporting: Tasks, inspections, and audits are executed within defined time windows, locations, roles, and approval flows to ensure accuracy and accountability.
- Physical-to-digital traceability: Operational activities are linked to real-world locations, assets, equipment, and environments through mechanisms such as QR codes, sensors, timestamps, and evidence capture.
- Structured non-conformance handling: Deviations are not treated as standalone events but flow into a controlled Corrective and Preventive Action (CAPA) lifecycle with severity, ownership, SLA, verification, and audit trails.
- Role-based simplicity with enterprise depth: Frontline users interact with a simplified, task-focused mobile experience, while supervisors and administrators manage configuration, automation, approvals, and analytics through advanced controls.
- Industry-configurable, single-core platform: The platform maintains one unified core architecture while adapting to different industries through configuration layers, templates, hierarchies, workflows, and automation rules—rather than custom code per industry.

This positioning directly addresses the need identified in the client review to move beyond a feature-centric description and present the platform as an enterprise-grade system for operational discipline, compliance enforcement, and continuous improvement.

All subsequent sections of this Scope of Work—including Industry Pack Architecture, QR enforcement, automation rules, manufacturing structures, and reporting—should be interpreted within this system philosophy.

3. Platform Scope Overview

The scope of this project covers the development of a modular, multi-tenant SaaS platform consisting of administrative configuration layers, company-level management tools, and execution-focused interfaces for frontline and managerial users. The platform is designed with a mobile-first execution model for operational staff and a web-based administrative and analytical interface for supervisors, managers, and system administrators.

3.1 Scope of the Platform

The platform scope includes the following major capability areas:

A. Centralized Administration & Multi-Tenant Management

- Platform-level administration for managing companies, subscriptions, modules, and system-wide settings
- Company onboarding, subscription configuration, and module entitlement management
- Centralized user management, role configuration, and access control
- Secure, scalable architecture supporting multiple organizations on a single platform

B. Company & Location-Based Operations Management

- Company-level dashboards providing real-time visibility into operational and compliance status
- Management of multiple locations, branches, sites, or facilities under a single organization
- Assignment of users, roles, templates, sensors, and devices at location level
- Location-specific configurations including schedules, operational hours, and access controls

C. Routine Execution Engine

- Creation, scheduling, assignment, and execution of:
- Checklists and operational forms
- Inspections and structured audits
- Shift-based and time-bound routines
- Support for daily, weekly, monthly, ad-hoc, and event-driven routines
- Evidence capture including photos, signatures, timestamps, and geolocation
- Offline-capable mobile execution for frontline staff

D. Audits, Inspections & Compliance Verification

- Configurable audit templates with scoring, weighted sections, and conditional logic
- Scheduled and ad-hoc audits across locations and assets
- Multi-level review and approval workflows
- Full audit history, reporting, and traceability

E. Corrective & Preventive Action (CAPA) Management

- Automatic and manual creation of CAPAs from checklists, audits, sensors, and incidents
- Severity-based prioritization and SLA-driven due dates
- Root cause analysis, task management, evidence attachment, and verification workflows
- Escalation and tracking of overdue or critical corrective actions

F. Sensors, IoT & Environmental Monitoring

- Integration and management of sensors (e.g., temperature, humidity, door, equipment status)
- Threshold-based alerts and breach detection
- Real-time monitoring dashboards and historical data analysis
- Sensor-triggered automation and CAPA generation

G. Automation & Alerts

- Rule-based automation engine enabling trigger-condition-action workflows
- Automated notifications, task assignments, escalations, and CAPA creation
- Reduction of manual follow-ups and supervisory overhead

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- Full execution logs and auditability of automated actions

H. Workforce Scheduling & Time Tracking

- Staff scheduling, shift management, and availability tracking
- Time clock functionality with PIN, QR, or kiosk-based authentication
- Attendance monitoring, overtime rules, and exception handling

I. Document & SOP Management

- Centralized document repository for SOPs, policies, manuals, and regulatory documents
- Version control, review cycles, and mandatory acknowledgements
- Role- and location-based document visibility and access control

J. Reporting & Analytics

- Operational and compliance dashboards for different user roles
- Standard reports covering checklists, audits, CAPAs, sensors, users, and locations
- Exportable reports (PDF, CSV, Excel) and scheduled report delivery
- Foundation for future business intelligence and advanced analytics

3.2 Scope Boundaries

The scope of this project includes platform development and configuration as described above. The following are considered out of scope unless explicitly agreed in writing:

- Industry-specific regulatory certification or legal advisory services
- Third-party ERP, POS, or HR system integrations beyond defined interfaces
- Custom hardware development (sensors, devices)
- On-site operational consulting or compliance audits

3.3 Platform Evolution

The platform scope is intentionally structured to support phased evolution:

- Phase-1 focuses on execution, enforcement, adoption, and operational stability
- Phase-2 introduces advanced intelligence layers such as business intelligence, AI-assisted insights, and predictive analytics

This phased approach ensures controlled delivery, faster adoption, and reduced implementation risk for enterprise customers.

4. Industry Pack Architecture

4.1 Purpose of Industry Pack Architecture

The Industry Pack Architecture defines how the platform adapts to different industries while maintaining a single, unified core system.

Although the platform operates as one multi-tenant SaaS solution, operational compliance requirements vary significantly across industries due to differences in:

- Asset and location hierarchies
- Operational routines and frequencies
- Compliance standards and inspection depth
- Risk severity models and escalation logic
- Reporting KPIs and audit expectations

The Industry Pack Architecture introduces a structured abstraction layer that enables the platform to support industry-specific operational models through configuration, templates, and rules—without requiring separate products or custom codebases. This approach directly addresses the requirement identified during the SOW review to move from a generic, feature-driven system to an enterprise-grade, industry-aware operational compliance platform.

4.2 What Is an Industry Pack

An Industry Pack is a predefined configuration layer that customizes how the core platform behaves for a specific industry.

Each Industry Pack may include:

- Predefined asset and location hierarchy models
- Industry-specific checklist, audit, and CAPA templates
- Default operational routines and schedules
- Role presets aligned with industry workflows

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- Automation and escalation rules appropriate to industry risk levels
 - Industry-relevant KPIs and reporting views

Industry Packs do not change the underlying platform modules. They configure how those modules are used, presented, and enforced.

4.3 Relationship Between Industry Packs and Core Platform

The platform core remains consistent across all industries and includes:

- Admin Panel
- Company Admin Panel
- Staff & Manager Execution Panels
- Core modules such as Checklists, Audits, CAPA, Sensors, Automation, Reporting, Documents, Scheduling, and Logbook

Industry Packs operate above these components as a configuration and behavior layer.

This ensures:

- Faster onboarding for new companies
- Consistent best practices per industry
- Reduced configuration errors
- Predictable compliance behavior across locations

4.4 Industry Pack Configuration Scope: Each Industry Pack may define or influence the following aspects of the platform:

4.4.1 Hierarchy & Asset Model

Defines the structural hierarchy used within the Company and Location Management sections.

Examples:

- Manufacturing: Factory → Production Line → Machine → Sensor
- Hospitality: Hotel → Floor → Room → Asset
- Food & Beverage: Outlet → Kitchen Area → Equipment → Sensor

These hierarchies are implemented using the existing Location, Zone, and Asset constructs described later in this SoW.

4.4.2 Routine & Template Presets

Industry Packs may preload:

- Checklist templates
- Audit templates
- CAPA templates
- Incident and logbook templates

Templates are tagged and categorized by industry and can be modified or extended by Company Admins as required.

4.4.3 Role & Responsibility Mapping

Industry Packs may provide default role mappings such as:

- Operator / Staff
- Supervisor
- Manager
- Quality or Compliance Lead

These role mappings align with the existing Roles & Privileges system and do not restrict custom role creation.

4.4.4 Automation & Escalation Defaults

Each Industry Pack may define default automation behaviors, including:

- Auto-generation of CAPAs on failures
- Severity thresholds and SLA expectations
- Escalation paths based on role and risk level
- Alerting behaviors for missed routines or sensor breaches

All automation rules remain editable through the Automation & Alerts Module.

4.4.5 Reporting & KPI Focus

Packs influence how reporting is consumed by emphasizing industry-relevant metrics, such as:

- Compliance scores
- Repeat failures
- Asset downtime

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- CAPA aging
 - Audit readiness

The reporting engine itself remains unchanged and configurable.

4.5 Initial Industry Packs (Phase-1)

Based on the current scope and client requirements, the following Industry Packs are included as part of Phase-1 architecture readiness:

- Manufacturing Industry Pack (first-class priority)
- Food & Beverage Industry Pack
- Hospitality Industry Pack
- Facility / Operations Management Industry Pack

Additional Industry Packs may be introduced in future phases through configuration and template expansion.

4.6 Custom Industry Configuration

In addition to predefined Industry Packs, the platform supports custom industry configurations, allowing organizations to:

- Modify hierarchy structures
- Create industry-specific templates
- Adjust automation rules
- Define custom KPIs

This ensures flexibility without compromising the integrity of the core platform.

4.7 Industry Pack Governance

Industry Packs are managed at the Admin Panel level and applied during:

- Company onboarding
- Subscription configuration
- Module entitlement setup

Company Admins may further customize industry presets within the boundaries of their assigned roles and permissions.

4.8 Architectural Outcome

By introducing Industry Pack Architecture, the platform achieves:

- Clear separation between core system capabilities and industry-specific behavior
- Scalability across multiple industries without product fragmentation
- Faster enterprise onboarding with reduced configuration overhead
- A foundation for advanced features such as QR enforcement, manufacturing hierarchies, and industry-aware automation

This architecture directly supports the enterprise requirements identified in the SOW review and enables the platform to evolve beyond a generic quality control system into a multi-industry operational compliance platform.

4.9 Industry Pack Differentiation at Execution Level

While the platform is built on a common, industry-agnostic core, Industry Packs introduce concrete operational differentiation at the execution level. Each Industry Pack defines how routines are performed, what metrics matter, how enforcement is applied, and how automation behaves for that specific operational context. The following examples illustrate how Industry Packs differ in practice.

4.9.1 Manufacturing Industry Pack — Execution Characteristics

In manufacturing environments, the platform operates with a production-centric execution model, focused on asset-level quality, continuity, and traceability.

Typical characteristics include:

- Operational Hierarchy
 - Factory → Production Line → Machine / Equipment
- Routine Execution
 - Shift start and shift end quality checks
 - In-shift machine inspections
 - Preventive maintenance routines
- Enforcement
 - Machine-level QR code enforcement
 - Inspections must be executed at the physical machine
- Automation & Quality Control
 - Inspection failure triggers automatic CAPA creation

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- Repeated failures escalate severity and notify supervisors
 - Maintenance issues can block further execution until resolved
 - Key Metrics
 - Machine compliance rate
 - Defect frequency
 - CAPA aging and recurrence
 - Downtime-related quality events

This pack supports continuous production environments where quality, uptime, and traceability are critical.

4.10 Industry Pack Behavioral Governance Model

Industry Packs function as behavioral governance layers within the platform, defining not only structural configuration but also execution logic, enforcement rules, automation behavior, and compliance interpretation for their respective operational contexts. While the platform core remains industry-agnostic, active Industry Packs govern how system mechanisms operate within a given environment.

4.10.1 Execution & Enforcement Alignment

Each Industry Pack may define:

- Applicable routine execution models
- Enforcement strictness and validation controls
- Asset or location binding requirements

This ensures that operational workflows align with industry-specific realities.

4.10.2 Automation & Rule Behavior

Automation logic is evaluated within the context of the active Industry Pack.

Industry Packs may influence:

- Trigger relevance and interpretation
- Corrective action behavior
- Escalation sensitivity and severity mapping

This prevents uniform automation behavior across fundamentally different operational environments.

4.10.3 QR & Validation Behavior

QR enforcement policies may vary by Industry Pack.

Examples include:

- Machine-centric enforcement (Manufacturing)
- Area- or equipment-centric enforcement (Food & Beverage)
- Location-centric enforcement (Facilities & Operations)

4.10.4 Metrics & Reporting Context

Industry Packs determine the operational significance of metrics and KPIs.

Reporting emphasis, thresholds, and compliance indicators may therefore differ across packs, ensuring that performance insights remain contextually meaningful. Industry Packs therefore operate as operational logic containers, ensuring that workflows, controls, and analytics remain aligned with the selected industry model rather than behaving as generic configurations.

5. Additional Supported Industries

In addition to the explicitly defined Industry Packs, the platform is designed to support a wide range of operational environments through its industry-agnostic core architecture, configurable hierarchies, templates, automation rules, and enforcement mechanisms.

Based on the existing platform capabilities described in this Scope of Work, the system can be configured to support the following industries without requiring new core modules or architectural changes:

- Facility Management & Building Operations
(e.g., commercial buildings, residential complexes, campuses)
- Warehousing, Cold Storage & Logistics
(e.g., storage facilities, distribution centers, inventory handling operations)
- Utilities & Energy Operations
(e.g., power facilities, water treatment plants, infrastructure maintenance)
- Airports & Transportation Facilities
(e.g., terminals, support facilities, operational zones)
- Education Institutions

(e.g., schools, colleges, universities, campus facilities)

- Petrol Stations & Fuel Retail Operations
- Healthcare Support & Non-Clinical Operations

(e.g., facility hygiene, equipment checks, safety routines)

- Laboratories & Testing Facilities

(e.g., quality labs, research environments, compliance-driven testing areas)

Support for these industries is achieved through:

- Configurable location, zone, and asset hierarchies
- Customizable checklist, audit, and CAPA templates
- QR-based execution and enforcement
- Rule-based automation and escalation
- Role-based access and reporting

Where required, industry-specific templates, KPIs, and automation rules may be introduced through configuration or future Industry Packs without altering the core platform architecture.

6. Manufacturing Industry Pack

6.1 Purpose of the Manufacturing Industry Pack

The Manufacturing Industry Pack defines how the platform supports factory-level operations, quality assurance, safety compliance, and equipment discipline within manufacturing environments. Manufacturing operations involve complex asset hierarchies, shift-based execution, machine-level inspections, preventive maintenance, and strict audit readiness. While the core platform modules (Checklists, Audits, CAPA, Sensors, Automation, Reporting, Documents, and Scheduling) are industry-agnostic, manufacturing requires a clearly defined operational model to ensure correct configuration, enforcement, and adoption. This section establishes manufacturing as a first-class industry, not a generic use case, in alignment with enterprise manufacturing expectations and the client's review feedback.

6.2 Manufacturing Hierarchy Structure

Within the Manufacturing Industry Pack, the platform supports the following logical hierarchy using the existing Location, Zone, and Asset constructs:

- Company
 - Factory
 - Production Line
 - Machine / Equipment
 - Sensors (optional)
 - Assigned Checklists and Audits

This hierarchy enables manufacturing organizations to manage compliance and quality controls at the appropriate operational level, from factory-wide audits down to individual machine inspections.

6.3 Manufacturing Routine Execution

6.3.1 Shift-Based Operations

The Manufacturing Industry Pack supports shift-based operational routines, including but not limited to:

- Line start-up inspections (per shift)
- Line shutdown inspections
- In-process quality checks
- Daily safety inspections
- Weekly preventive maintenance checks
- Monthly compliance and quality audits

Routines may be scheduled, assigned, and enforced using the existing Checklist, Audit, Scheduling, and Automation modules.

6.3.2 Machine-Level Execution

Manufacturing routines may be associated directly with:

- Production lines
- Individual machines or equipment
- Supporting areas such as utilities, warehouses, or QA labs

This enables machine-specific execution, traceability, and historical analysis.

6.4 Manufacturing Quality & Compliance Audits

The Manufacturing Industry Pack supports structured audits such as:

- Safety audits
- Quality audits (e.g., ISO readiness, internal GMP checks)
- 5S audits
- Environmental and facility audits

Audits may include:

- Weighted scoring
- Evidence capture (photos, comments, signatures)
- Multi-level review and approval workflows
- Automatic generation of non-conformances and CAPAs

6.5 Manufacturing CAPA Lifecycle

Manufacturing-related non-conformances may originate from:

- Failed checklist items
- Audit findings
- Sensor threshold breaches
- Incident reports

The platform supports a full CAPA lifecycle for manufacturing environments, including:

- Severity classification (Low, Medium, High, Critical)
- SLA-driven due dates
- Task-level corrective and preventive actions
- Root cause analysis
- Verification and closure workflows
- Escalation for overdue or critical CAPAs

6.6 Sensors & Preventive Maintenance in Manufacturing

The Manufacturing Industry Pack leverages the existing Sensors & IoT module to support:

- Equipment condition monitoring
- Environmental compliance (temperature, humidity)
- Machine calibration tracking

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- Preventive maintenance reminders
 - Sensor-triggered alerts and CAPA creation

Sensors may be associated with specific machines, production lines, or factory zones.

6.7 Manufacturing Reporting & KPIs

The Manufacturing Industry Pack emphasizes reporting and analytics relevant to manufacturing operations, including:

- Line-level compliance scores
- Machine inspection completion rates
- Defect and non-conformance frequency
- CAPA aging and closure rates
- Preventive vs corrective maintenance ratios
- Audit readiness and trend analysis

These metrics are generated using the existing Reporting & Analytics module and may be filtered by factory, line, machine, shift, or time period.

6.8 Role Alignment for Manufacturing Operations

The Manufacturing Industry Pack aligns with the platform's role-based access model and may include default role mappings such as:

- Operator / Line Staff
- Supervisor
- Maintenance Technician
- Quality / Compliance Manager
- Factory or Operations Manager

Role mappings are configurable through the existing Roles & Privileges system and do not restrict custom role definitions.

6.9 Manufacturing Industry Pack Scope

The Manufacturing Industry Pack does not introduce new modules. It configures and orchestrates existing platform capabilities to meet manufacturing operational requirements through:

- Defined hierarchies

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- Routine templates
 - Automation defaults
 - Reporting focus
 - Enforcement mechanisms (described in subsequent sections)

This approach ensures scalability, maintainability, and consistency across industries while meeting manufacturing-specific needs.

6.10 Manufacturing Operational Model (Execution & Control)

Within the Manufacturing Industry Pack, the platform operates according to a shift-driven, asset-centric operational model, designed to support continuous production environments where quality, machine condition, and execution traceability are critical.

Operational behavior within this pack includes:

6.10.1 Shift-Based Execution Framework

Manufacturing routines are structured around production shifts.

Typical execution patterns include:

- Shift Start Controls
 - Mandatory machine readiness and quality checks
 - Verification of safety and operational conditions
 - Enforcement via machine-level QR codes where configured
- In-Shift Operational Controls
 - Periodic or event-driven machine inspections
 - Quality control checklists tied to specific equipment
 - Immediate logging of anomalies or deviations
- Shift End Controls
 - Machine condition verification
 - Production integrity and compliance confirmation
 - Execution is time-bound and operator-linked, preventing retrospective or out-of-context submissions.

6.10.2 Machine-Centric Enforcement

All quality and inspection routines may be bound to specific machines or equipment.

Where QR enforcement is enabled:

- Inspections must be executed at the physical machine
- Execution without valid scan is not permitted
- Each record is traceable to machine, operator, and timestamp

6.10.3 Quality & Maintenance Interaction

The platform supports operational linkage between:

- Machine inspections
- Preventive maintenance routines
- Corrective actions (CAPA)

Typical system behaviors include:

- Inspection failure → automatic CAPA initiation
- Repeated failures → severity escalation
- Maintenance-related issues → may block further execution

6.10.4 Manufacturing-Relevant Metrics

Reporting and analytics within this pack prioritize:

- Machine compliance rates
- Defect and non-conformance frequency
- CAPA aging and recurrence patterns
- Downtime-associated quality events

These metrics support production stability, quality assurance, and supervisory oversight.

6.11 Architectural Outcome

By introducing the Manufacturing Industry Pack, the platform:

- Explicitly supports factory, line, and machine-level operations
- Enables shift-based execution and enforcement
- Improves audit readiness and traceability
- Reduces operational risk through automation and CAPA discipline
- Aligns the platform with enterprise manufacturing expectations

This section directly addresses the manufacturing-related gaps identified during the SOW review and establishes a solid foundation for QR-based enforcement and automation workflows.

7. QR Enforcement Lifecycle

7.1 Purpose of QR Enforcement

The QR Enforcement Lifecycle defines how the platform ensures physical presence, correct asset context, and accurate execution for all operational routines, inspections, and audits. In environments such as manufacturing, food & beverage, hospitality, and facilities management, compliance failures often occur due to:

- Remote or falsified checklist entries
- Incorrect asset or location selection
- Delayed or skipped inspections

The QR Enforcement Lifecycle introduces a mandatory physical-to-digital control layer that binds inspections and routines to real-world locations, assets, and equipment, ensuring authenticity, traceability, and enforcement.

7.2 QR Code Scope and Coverage

QR codes may be generated and assigned at multiple operational levels, including but not limited to:

- Factories
- Production lines
- Machines and equipment
- Rooms and zones
- Utilities and shared assets
- Vehicles or mobile assets (where applicable)

Each QR code uniquely represents a specific physical entity within the platform hierarchy.

7.3 QR Code Generation and Assignment

QR codes are generated and managed through the Admin Panel or Company Admin Panel and are associated with:

- Company identifier

- Location or factory identifier
- Zone, line, or asset identifier
- Entity type (e.g., factory, line, machine, room)

QR codes may be:

- Printed and affixed to physical assets or locations
- Embedded in labels generated through the Labeling module
- Replaced or deactivated if assets are retired or relocated

All QR assignments are logged for auditability.

7.4 QR Scan Validation Workflow

When a user scans a QR code using the mobile application, the system enforces the following validation sequence:

- User Authentication
 - Confirms the user is logged in and active
 - Validates role and permissions
- Entity Validation
 - Confirms the QR code corresponds to an active location, line, or asset
 - Ensures the entity is assigned to the user's organization
- Role Authorization
 - Confirms the user is authorized to perform actions on the scanned entity
- Time and Schedule Validation
 - Confirms the scan occurs within the permitted execution window
 - Validates shift timing or scheduled routine requirements
- Contextual Task Resolution
 - Automatically opens the correct checklist, inspection, or audit
 - Manual selection of location or asset is restricted where QR enforcement is enabled

If any validation fails, execution is blocked and logged.

7.5 QR Enforcement & Validation Controls

QR codes within the platform operate as enforcement controls, not merely identification or navigation mechanisms. Where QR enforcement is enabled for a routine, asset, or location:

7.5.1 Execution Authorization

- A valid QR scan is mandatory prior to execution
- Execution without successful validation is not permitted
- Manual bypass or retrospective submission is disallowed

7.5.2 Context Binding

Each QR-enforced execution is bound to:

- The associated asset or location
- The authenticated user / operator
- The execution timestamp

This prevents out-of-context or unauthorized submissions.

7.5.3 Anti-Fraud Safeguards

The platform validates QR scans against:

- Authorized user roles
- Permitted execution windows (where configured)
- Assigned assets or operational zones

Invalid or unauthorized scans may result in:

- Rejection of execution attempt
- Logging of enforcement violation
- Triggering of supervisory alerts (where applicable)

7.5.4 Compliance Integrity

QR enforcement ensures that:

- Inspections occur at the intended physical context
- Records remain traceable and auditable
- False or remote compliance reporting is minimized

QR codes therefore act as operational gatekeepers within controlled workflows.

7.6 Enforced Execution Behavior

When QR enforcement is enabled:

- Users cannot submit checklists or inspections without scanning the assigned QR code

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- Manual location or asset selection is disabled for enforced routines
 - Each submission is automatically linked to:
 - Location or asset
 - User
 - Timestamp
 - Evidence (photos, signatures, sensor data where applicable)

This ensures all execution data is tamper-resistant and traceable.

7.7 Manufacturing-Specific QR Enforcement

Within the Manufacturing Industry Pack, QR enforcement supports:

- Machine-level inspections (QR on machine opens only that machine's checklist)
- Line-level start-up and shutdown routines
- Shift-based enforcement (QR scan required per shift)
- Maintenance and calibration workflows tied to specific equipment

This ensures that manufacturing routines are executed at the correct physical point of operation.

7.8 QR Enforcement and Automation Integration

QR-based execution events may trigger automation rules, including:

- Auto-creation of CAPAs for failed inspections
- Alerts for missed or late QR scans
- Escalation when required QR-based routines are not completed
- Sensor validation against scanned assets

All automation behavior is managed through the Automation & Alerts module.

7.9 Anti-Fraud and Control Measures

The QR Enforcement Lifecycle includes safeguards to prevent misuse, including:

- Prevention of duplicate or rapid repeated scans
- Validation against inactive or relocated assets
- Blocking execution from unauthorized devices or users
- Logging of all scan attempts, including failed validations

These controls support audit readiness and regulatory compliance.

7.10 Reporting and Audit Traceability

QR enforcement data is available within the Reporting & Analytics module, enabling visibility into:

- QR-based execution compliance rates
- Missed or failed QR-enforced routines
- Asset-level inspection history
- User execution patterns and deviations

QR scan records form part of the platform's immutable audit trail.

7.11 QR Enforcement Scope and Configuration

QR enforcement rules may be configured at multiple levels:

- Industry Pack defaults
- Company-level settings
- Location, line, or asset-level overrides

Company Admins may enable or disable QR enforcement per routine, inspection, or audit based on operational requirements.

7.12 Architectural Outcome

The QR Enforcement Lifecycle ensures that the platform moves beyond self-reported compliance to verified operational execution. By binding routines to physical assets and locations, the platform:

- Eliminates false or remote entries
- Improves data accuracy and trust
- Strengthens audit defensibility
- Enables automation and escalation based on real execution events

This enforcement layer is a core component of the platform's enterprise-grade operational compliance architecture.

8. Automation Rule Engine (Trigger → Action)

8.1 Purpose of the Automation Rule Engine

The Automation Rule Engine defines how the platform reduces manual supervision, enforces accountability, and ensures timely corrective action through event-driven automation. In operational environments, compliance failures often persist due to:

- Missed inspections not being followed up
- Delayed corrective actions
- Manual escalation dependent on supervisors
- Inconsistent enforcement across locations

The Automation Rule Engine introduces a structured Trigger → Condition → Action framework that enables the platform to automatically respond to operational events in real time. This engine is a core component of the platform's enterprise-grade operational compliance architecture.

8.2 Automation Architecture Overview

Automation rules are configured and managed through the Automation & Alerts Module and operate across all core platform modules, including:

- Checklists & Forms
- Inspections & Audits
- Corrective & Preventive Actions (CAPA)
- Sensors & IoT
- QR Enforcement events
- Scheduling & Time Clock
- Documents & Acknowledgements
- Logbook & Incident Reports

Each automation rule follows a standardized structure:

Trigger → Conditions → Actions

8.3 Supported Automation Triggers

Automation triggers represent system events that initiate rule execution. Supported trigger categories include, but are not limited to:

8.3.1 Checklist Triggers

- Checklist completed
- Checklist failed

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- One or more questions failed
 - Checklist missed
 - Checklist overdue

8.3.2 Audit Triggers

- Audit completed
- Audit score below defined threshold
- Non-conformance recorded
- Audit overdue

8.3.3 CAPA Triggers

- CAPA created
- CAPA overdue
- CAPA moved to a new stage
- CAPA closed or reopened

8.3.4 Sensor Triggers

- Sensor threshold breach (temperature, humidity, etc.)
- Sensor offline
- Low battery alert
- Calibration due or overdue

8.3.5 QR Enforcement Triggers

- Required QR scan not performed
- QR-enforced routine missed
- QR scan attempted by unauthorized user
- QR scan outside permitted time window

8.3.6 Scheduling & Workforce Triggers

- Late clock-in
- Missed shift
- Overtime triggered

8.3.7 Document & Logbook Triggers

- New document published
- Document acknowledgement overdue
- Incident report created

8.4 Conditions and Rule Logic

Each automation trigger may be refined using one or more conditions to control when actions are executed. Examples of supported conditions include:

- Location, factory, line, or asset
- User or role
- Severity level
- Score thresholds
- Repetition count (e.g., same failure multiple times within a period)
- Time-based conditions (e.g., duration exceeded)

Conditions may be combined using AND / OR logic, including nested condition groups.

8.4.1 Deterministic Automation Enforcement

Automation rules within the platform operate as deterministic control mechanisms, ensuring that defined operational events produce non-optional system reactions.

Where automation rules are configured and active:

- Trigger conditions result in mandatory execution of associated actions
- System reactions are not advisory or discretionary
- User intervention is not required to initiate defined corrective flows

Typical deterministic behaviors include:

- Checklist or Audit Failure
 - Automatically initiates corrective workflows, including CAPA generation where applicable
- Repeated Non-Conformance
 - Escalates severity levels based on predefined rule logic
 - Triggers supervisory or managerial notification sequences
- Overdue Tasks or CAPAs
 - Generates escalation events according to the Severity and SLA Model
 - May require higher-level acknowledgment or intervention
- Threshold or Condition Violations
 - Initiates predefined enforcement, notification, or assignment actions

Deterministic automation ensures:

- Silent failures are prevented

-
- Compliance risks are surfaced without manual oversight dependency
 - Operational responses remain consistent, traceable, and auditable

Automation therefore functions as an active operational control layer, rather than a passive notification system.

8.5 Supported Automation Actions

When trigger and condition criteria are met, the platform may execute one or more actions automatically.

8.5.1 Notification Actions

- Send in-app notifications
- Send email alerts
- Notify specific users, roles, or managers
- Send escalation notifications

8.5.2 CAPA Actions

- Automatically create a CAPA
- Assign CAPA to a user or role
- Set severity and due date based on rules
- Escalate CAPA if overdue

8.5.3 Task and Assignment Actions

- Assign follow-up checklists or inspections
- Reassign missed or overdue routines
- Create corrective tasks linked to CAPAs

8.5.4 Escalation Actions

- Escalate issues to supervisors or managers
- Increase severity level after repeated failures
- Trigger secondary actions if primary actions are ignored

8.5.5 System Actions

- Change CAPA stage
- Flag entries for review
- Log enforcement events for audit tracking

8.6 Manufacturing-Specific Automation

Within the Manufacturing Industry Pack, automation rules may include:

- Auto-creation of CAPAs for machine-level checklist failures
- Escalation of repeated line start-up failures
- Immediate alerts for sensor breaches on critical equipment
- Preventive maintenance reminders based on time or sensor data
- Escalation for missed QR-enforced shift routines

These rules ensure consistent enforcement across factories, production lines, and machines.

8.7 Severity and SLA Model

The Automation Rule Engine supports severity-based handling of non-conformances.

Example severity model:

- Low – Informational, non-urgent (longer SLA)
- Medium – Requires timely correction
- High – Operational risk, short SLA
- Critical – Immediate attention required

Severity levels influence:

- CAPA due dates
- Escalation timing
- Notification priority

8.8 Automation Governance and Safety Controls

To prevent misuse or unintended behavior, the Automation Rule Engine includes governance controls such as:

- Role-based permissions for rule creation and editing
- Conflict detection between automation rules
- Prevention of infinite automation loops
- Execution logs for all automation events
- Ability to activate, deactivate, or audit rules

8.9 Reporting Architecture & Traceability Model

Reporting within the platform functions as an operational intelligence and compliance traceability mechanism, ensuring that automation events, enforcement actions, and operational outcomes remain observable, verifiable, and decision-relevant.

8.9.1 Automation-Derived Visibility

Reporting outputs are directly linked to system events, including:

- Triggered automation rules
- Corrective and escalation actions
- Compliance-impacting conditions

This ensures that reported metrics reflect validated operational behavior, not manual interpretation.

8.9.2 Traceability & Drill-Down Structure

All reported indicators support logical traceability:

Metric / KPI → Operational Entity → Event / Record → Evidence

Example:

Compliance Deviation → Asset / Machine → Trigger Event → Execution Record

Aggregated metrics are therefore grounded in auditable system records.

8.9.3 Corrective Action & Compliance Insights

Given the platform's enforcement-driven design, reporting prioritizes:

- CAPA lifecycle and aging
- Recurring trigger patterns
- Threshold and violation events
- Resolution and SLA adherence

8.9.4 Audit & Governance Integrity

Reporting data is derived exclusively from system-validated actions, including:

- QR-enforced executions
- Deterministic automation reactions
- User-attributed operational records

This ensures reliability for operational oversight, compliance verification, and management review contexts.

8.10 Reporting and Auditability

All automation executions are recorded and available for review, including:

- Trigger event details
- Conditions evaluated
- Actions executed
- Execution outcome (success or failure)

Automation history forms part of the platform's audit trail and supports compliance verification.

8.11 Architectural Outcome

The Automation Rule Engine transforms the platform from a manual oversight system into an event-driven operational enforcement engine. By automating responses to failures, delays, and risks, the platform:

- Reduces dependency on manual supervision
- Improves response times
- Ensures consistent enforcement across locations
- Strengthens compliance and audit readiness

This automation layer is foundational to enterprise scalability and operational reliability.

9. Phase-1 vs Phase-2 Roadmap

9.1 Purpose of the Phased Roadmap

This section defines a clear separation between Phase-1 (Execution & Adoption) and Phase-2 (Intelligence & Optimization) to ensure controlled delivery, predictable timelines, and enterprise-grade scalability. The phased approach addresses the following objectives:

- Prevent scope creep during initial implementation
- Prioritize operational execution and user adoption
- Establish a stable data foundation before advanced intelligence layers
- Clearly position advanced analytics and AI capabilities as a subsequent phase

This separation is critical for enterprise rollout, vendor accountability, and stakeholder alignment.

9.2 Phase-1: Execution & Operational Enforcement

9.2.1 Phase-1 Objective

Phase-1 focuses on delivering a fully functional operational compliance execution platform that enforces routines, captures evidence, automates corrective actions, and provides real-time visibility across locations and assets.

The emphasis in Phase-1 is execution accuracy, adoption, and reliability, not predictive intelligence.

8.2.2 Phase-1 Scope (Included)

Phase-1 includes the implementation and stabilization of the following capabilities:

- Core Platform & Panels
 - Admin Panel
 - Company Admin Panel
 - Staff & Manager Execution Panels
 - Multi-tenant SaaS architecture
 - Role-based access and permissions
- Industry & Architecture
 - Industry Pack Architecture
 - Manufacturing Industry Pack (first-class support)
 - Industry-aware hierarchy models (factory, line, machine, asset)
- Routine Execution
 - Checklists and forms
 - Inspections and audits
 - Shift-based and scheduled routines
 - Evidence capture (photos, signatures, timestamps)
 - Offline-capable mobile execution
- QR Enforcement
 - QR generation and assignment
 - Location and asset-locked execution
 - Role and time-based validation
 - Anti-fraud controls and scan audit logs
- Automation & CAPA
 - Trigger → Condition → Action automation rules

-
- Automatic CAPA creation
 - Severity-based SLAs and escalation
 - Task tracking, verification, and closure workflows
 - Sensors & Monitoring
 - Sensor registration and configuration
 - Threshold alerts
 - Sensor-triggered automation and CAPAs
 - Calibration and maintenance tracking
 - Operational Reporting
 - Real-time dashboards
 - Compliance and completion metrics
 - CAPA status and aging reports
 - Audit results and trends
 - Exportable reports (PDF, CSV, Excel)

9.2.3 Phase-1 Outcome

At the completion of Phase-1, the platform will function as a reliable, enforceable operational compliance system capable of:

- Preventing false or remote data entry
- Enforcing correct execution at physical locations and assets
- Automatically responding to failures and risks
- Providing auditable, regulator-ready records

Phase-1 establishes the data integrity and operational discipline required for advanced intelligence.

9.3 Phase-2: Intelligence, Optimization & AI Assistance

9.3.1 Phase-2 Objective

Phase-2 builds upon the stable execution foundation of Phase-1 to introduce advanced analytics, insights, and controlled AI-assisted capabilities.

Phase-2 focuses on decision support and optimization, not execution control.

9.3.2 Phase-2 Scope (Future Enhancements) Phase-2 may include, subject to separate approval and planning:

- Business Intelligence & Analytics
 - Compliance trend analysis
 - Risk heatmaps by location, asset, or process
 - Cross-location performance benchmarking
 - Asset and routine performance insights
- Predictive & Advanced Insights
 - Early warning indicators for declining compliance
 - CAPA recurrence analysis
 - Asset-level risk scoring
 - Preventive maintenance optimization
- AI-Assisted Capabilities (Controlled Mode)
 - AI-generated audit summaries
 - Suggested root cause insights (read-only)
 - Compliance trend explanations
 - Executive-level digest reports
- AI capabilities in Phase-2 will:
 - Operate in advisory / read-only mode
 - Not perform autonomous actions
 - Be fully auditable and permission-controlled
 - Respect enterprise security and data governance policies

9.4 Explicit Phase Boundaries

To avoid ambiguity:

- Phase-1 does not include predictive analytics, autonomous AI actions, or external BI integrations beyond defined reports.
- Phase-2 does not modify execution logic without explicit approval.

Any Phase-2 features require:

- Separate scope definition
- Timeline estimation

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- Commercial agreement

9.5 Roadmap Governance The phased roadmap enables:

- Controlled delivery milestones
- Clear vendor accountability
- Predictable implementation timelines
- Enterprise-friendly adoption and rollout

This structure directly addresses the client's requirement for roadmap clarity and prevents over-engineering during initial deployment.

9.6 AI Agent Mode (Controlled Assistance Framework)

The platform may provide AI-assisted capabilities operating under a strictly controlled assistance framework, designed to preserve operational integrity, human authority, and auditability. AI functions exclusively as an advisory and assistive mechanism, not an autonomous decision-making or execution entity.

9.6.1 Human Authority & Decision Control

- AI-generated suggestions, analyses, or recommendations require human review
- No AI-driven action is executed without explicit user or system rule authorization
- Accountability for operational decisions remains with designated human roles

9.6.2 Operational Boundaries

AI mechanisms do not:

- Modify compliance records autonomously
- Execute workflows or corrective actions independently
- Bypass enforcement, validation, or approval controls

AI outputs are therefore informational and assistive in nature.

9.6.3 Traceability & Auditability

AI interactions, where applicable, may be:

- Logged within system audit trails
- Contextually associated with user sessions or workflows
- Subject to the same visibility and governance controls as other system events

9.6.4 Model & Vendor Neutrality

The AI assistance layer is designed to remain LLM-agnostic and vendor-neutral, allowing controlled evolution of AI technologies without altering core platform governance or operational logic.

AI Agent Mode is therefore positioned as a controlled augmentation layer, enhancing user productivity and decision support while preserving compliance, validation, and accountability structures.

9.7 Architectural Outcome

By separating execution and intelligence into distinct phases, the platform:

- Delivers immediate operational value in Phase-1
- Preserves long-term scalability and innovation in Phase-2
- Aligns with enterprise procurement and governance expectations
- Supports future AI and BI expansion without compromising core execution

10. Enterprise Non-Functional Requirements

10.1 Purpose

This section defines the non-functional requirements applicable to the entire platform, including all panels, modules, industry packs, enforcement layers, and automation workflows. These requirements specify how the system must operate, rather than what features it provides, and apply uniformly across Phase-1 deliverables unless otherwise stated.

10.2 Security & Access Control

10.2.1 Authentication

- Role-based authentication across all user types
- Support for PIN-based authentication for frontline users
- Session timeout and automatic logout for inactive sessions
- Secure password policies for administrative users

10.2.2 Authorization

- Fine-grained role-based access control (RBAC)
- Permissions enforceable at:
 - Company level

- Location / factory level
- Module level
- Asset level (where applicable)
- Users may only access data and actions explicitly permitted by assigned roles

10.2.3 Data Isolation

- Logical data separation between companies in a multi-tenant environment
- No cross-tenant data visibility or access under any circumstances

10.3 Performance & Responsiveness

- Mobile and web interfaces must support near-real-time response for routine operations
- Frontline execution actions (checklist open, submit, QR scan) should complete within acceptable operational latency under normal load
- Administrative dashboards must support responsive loading for enterprise datasets

10.4 Scalability & Availability

- The platform must support:
 - Multiple companies
 - Multiple locations per company
 - High numbers of concurrent frontline users
- Architecture must allow horizontal scaling to support growth in:
 - Users
 - Locations
 - Assets
 - Sensors
- Platform availability targets must support enterprise operational usage

10.5 Reliability & Fault Tolerance

- The system must be resilient to partial failures without data loss
- Failed automation executions must be logged and retried or flagged
- Sensor data interruptions must not corrupt historical records
- Graceful degradation must be supported where possible

10.6 Offline & Data Synchronization

- Mobile execution interfaces must support offline operation for:
 - Checklist execution

-
- Evidence capture
 - QR scans
 - Offline data must be securely stored on the device
 - Automatic synchronization must occur once connectivity is restored
 - Conflict resolution rules must prevent data corruption

10.7 Data Integrity & Validation

- Mandatory fields must be enforced during execution
- Invalid or incomplete submissions must be prevented
- QR-enforced routines must not allow bypass through manual entry
- Automation actions must execute only after successful validation

10.8 Compliance & Regulatory Readiness

- The platform must support audit-ready records suitable for internal and external compliance reviews
- Evidence, approvals, and corrective actions must be traceable end-to-end
- Data handling must support compliance with applicable enterprise data protection standards

10.9 Configuration Governance

- Only authorized users may:
 - Modify templates
 - Change automation rules
 - Alter industry pack configurations
- Configuration changes must be logged and auditable
- Critical configuration changes may require multi-level approval (where enabled)

10.10 Logging, Monitoring & Diagnostics

- System-level logging must be available for:
 - Errors
 - Automation failures
 - Integration issues
- Logs must support troubleshooting and support investigations
- Administrative users must have controlled access to system diagnostics

10.11 Phase Applicability

Unless explicitly stated otherwise:

- All Non-Functional Requirements in this section apply to Phase-1 deliverables
- Enhancements or extensions in Phase-2 must not degrade Phase-1 guarantees

10.12 Architectural Outcome

These Non-Functional Requirements ensure that the platform operates as an enterprise-grade, secure, scalable, and auditable operational compliance system, suitable for manufacturing and other regulated or high-risk industries.

11. Enterprise Service Reliability & Operational SLA Model

11.1 Purpose of Service Reliability Model

The Service Reliability & Operational SLA Model defines the platform's operational availability objectives, incident handling structure, and reliability governance principles required for enterprise-grade usage. Given that the platform functions as a mission-critical operational compliance and execution system, reliability controls are designed to ensure predictable system accessibility, controlled failure handling, and transparent operational recovery mechanisms.

11.2 Platform Availability Objectives

The platform is designed to support high-availability operational environments.

Subject to the agreed hosting model and infrastructure tier, the system targets enterprise-grade availability levels appropriate for routine execution, inspections, QR enforcement, automation workflows, and reporting access.

Availability objectives are governed by:

- Infrastructure redundancy and failover strategies
- Monitoring and alerting mechanisms
- Controlled maintenance procedures

Precise uptime commitments, where contractually required, may be defined under separate commercial or service agreements.

11.3 Incident Severity Classification Model

Operational events are categorized using a structured severity framework to ensure consistent response prioritization.

Typical severity categories include:

- P1 – Critical Service Impact
Complete or near-complete loss of core platform functionality affecting multiple users or locations.
- P2 – High Service Degradation
Major functional impairment with partial system availability.
- P3 – Limited / Localized Impact
Non-critical issues with limited operational disruption.

Severity classification governs response prioritization, escalation logic, and recovery workflows.

11.4 Incident Response & Recovery Principles

The platform's operational model follows controlled incident management principles:

- Continuous system monitoring and anomaly detection
- Prioritized response based on severity classification
- Root cause investigation and corrective stabilization
- Logging and auditability of incident events

Response and recovery objectives are influenced by:

- Hosting infrastructure characteristics
- Failure domain scope
- External dependency conditions

11.5 Planned Maintenance & Service Continuity

To preserve system stability and security posture, controlled maintenance activities may occur, including:

- Security updates
- Infrastructure upgrades
- Performance optimizations

Where feasible, maintenance operations are designed to minimize disruption to frontline execution workflows.

11.6 Reliability Boundaries & External Dependencies

Platform reliability is influenced by factors outside direct application control, including:

- Network connectivity
- Cloud infrastructure providers
- Third-party service dependencies

Service interruptions arising from such dependencies are managed through standard mitigation and recovery mechanisms but do not constitute application design failures.

11.7 Architectural Outcome

This Service Reliability Model ensures that the platform operates as an enterprise-ready operational system capable of supporting enforcement-driven workflows, automation logic, and distributed frontline execution with predictable reliability characteristics.

12. Enterprise Support & Incident Management Model

12.1 Purpose of Support & Incident Model

The Enterprise Support & Incident Management Model defines the structured mechanisms through which operational issues, system anomalies, and service disruptions are handled to ensure continuity of platform usage within enterprise environments. Given the platform's role in enforcement-driven workflows, inspections, automation logic, and compliance tracking, a controlled support and incident handling framework is maintained.

12.2 Support Interaction Channels

The platform is supported through designated communication and issue management channels appropriate for enterprise operations.

Support interactions may include:

- Issue reporting and tracking mechanisms
- Technical assistance and investigation workflows
- Operational guidance for platform usage and configuration

Available channels may vary depending on the agreed service model or commercial arrangement.

12.3 Incident Handling Framework

Reported incidents and detected operational anomalies are processed through a structured handling lifecycle:

- A. Incident Registration – Capture and classification of the issue
- B. Impact Assessment – Determination of severity and scope
- C. Response Initiation – Investigation and mitigation actions
- D. Resolution & Stabilization – Restoration of normal behavior
- E. Post-Incident Analysis – Root cause and corrective measures

This model ensures consistent handling of platform-affecting events.

12.4 Escalation & Priority Logic

Incidents are prioritized and escalated based on operational impact and severity classification.

Escalation logic may involve:

- Reclassification of incident severity
- Involvement of higher-level technical oversight
- Extended investigation or mitigation workflows

This prevents unresolved high-impact conditions from remaining unaddressed.

12.5 Maintenance & Update Governance

To preserve system stability, security posture, and performance characteristics, controlled platform updates and maintenance activities may be performed.

Such activities may include:

- Corrective patches
- Security updates
- Performance optimizations

Where feasible, maintenance activities are designed to minimize disruption to operational workflows.

12.6 Client-Side Responsibilities & Dependencies

Platform support effectiveness may depend on factors outside direct application control, including:

- Client network conditions
- User environment configurations
- Third-party service dependencies

Issues originating from such conditions may require cooperative investigation.

12.7 Operational Outcome

This support and incident management framework ensures that platform usage remains predictable, recoverable, and operationally sustainable within distributed and compliance-sensitive environments.

13. Security & Compliance Posture

13.1 Purpose of Security Posture

The platform is designed to operate within compliance-sensitive and operationally critical environments, where data integrity, execution reliability, and controlled access are essential.

The Security & Compliance Posture defines the governing principles for protecting platform data, ensuring tenant isolation, and maintaining operational trustworthiness.

13.2 Secure-by-Design Principles

Security controls are embedded at architectural and operational levels.

Core design principles include:

- Logical tenant isolation within a multi-tenant architecture
- Role-based access control mechanisms
- Controlled authentication and authorization flows
- Protection of data in transit and at rest (where applicable)

Security mechanisms are intended to reduce unauthorized access risks and preserve record integrity.

13.3 Data Protection & Integrity Model

The platform prioritizes the protection of operational and compliance records, including checklists, audits, CAPAs, automation events, QR-enforced executions, and associated evidence.

Protection mechanisms may include:

- Access-controlled data visibility
- Integrity-preserving record structures
- Controlled modification pathways
- System-level logging and traceability

These controls ensure that reported metrics and records remain defensible and auditable.

13.4 Compliance Alignment Philosophy

The platform's architecture and operational model are designed to support alignment with common enterprise security and compliance expectations.

Depending on deployment model and commercial agreements, controls may be configured to support organizational compliance requirements.

References to specific regulatory frameworks or certifications (e.g., SOC, ISO) represent design compatibility objectives, not implicit certification claims, unless explicitly stated under separate agreements.

13.5 Responsibility & Shared Security Considerations

Platform security operates within a shared responsibility model.

While the application enforces logical security controls and access governance, overall security posture may be influenced by:

- Hosting infrastructure characteristics
- Client-side identity management practices
- Network and endpoint security conditions

Certain security controls may therefore depend on cooperative operational practices.

13.6 Monitoring, Logging & Traceability

Security-relevant events and operational actions may be captured through platform logging and monitoring mechanisms.

Such events may include:

-
- Authentication and access events
 - Enforcement violations
 - Automation-triggered actions
 - System anomalies

This supports investigation, auditability, and controlled incident analysis.

13.7 Architectural Outcome

This Security & Compliance Posture ensures that the platform operates as a controlled, traceable, and enterprise-compatible system suitable for operational governance and compliance-driven workflows.

14. Identity & Access Management Model

14.1 Purpose of Identity & Access Model

The platform implements a controlled Identity & Access Management (IAM) framework to ensure that system access, operational permissions, and enforcement-driven workflows remain governed by authenticated user identity and defined authorization policies. This model supports secure interaction with compliance records, operational routines, automation mechanisms, and reporting outputs.

14.2 Authentication Framework

Platform access is governed through authenticated user identity.

Authentication mechanisms may include:

- Credential-based authentication
- Federated identity or Single Sign-On (SSO) models where supported
- Multi-factor authentication mechanisms where configured

Authentication flows are designed to prevent unauthorized system access while remaining compatible with enterprise identity ecosystems.

14.3 Authorization & Role Governance

Access privileges are governed using role-based access control principles.

User roles may define:

- Functional permissions (view, execute, approve, configure)
- Scope boundaries (location, asset, operational domain)
- Enforcement responsibilities (inspection, CAPA, audit actions)

Authorization controls ensure that system actions align with organizational responsibilities rather than unrestricted user access.

14.4 Enterprise Identity Integration Compatibility

The platform is designed to operate alongside enterprise identity providers and access governance systems.

Depending on deployment and integration requirements, the system may support:

- Federated authentication models
- Centralized identity validation workflows
- External identity lifecycle coordination mechanisms

References to identity standards or provisioning protocols represent architectural compatibility objectives, not implicit commitments, unless explicitly defined under separate technical agreements.

14.5 Accountability & Identity Traceability

All operational actions, including checklist execution, QR-enforced routines, automation-triggered events, and corrective workflows, remain attributable to authenticated user identity.

This ensures:

- Action traceability
- Compliance defensibility
- Audit and investigation integrity

14.6 Access Integrity & Control Boundaries

The IAM framework enforces:

- Prevention of unauthorized privilege escalation
- Controlled role assignment mechanisms
- Consistent permission evaluation across workflows

This maintains integrity across distributed operational environments.

14.7 Architectural Outcome

This Identity & Access Management Model ensures that platform usage remains securely governed, role-consistent, and enterprise-compatible without constraining organizational identity infrastructure choices.

15. Integration & API Strategy

15.1 Purpose of Integration Model

The platform is designed to operate within heterogeneous enterprise technology ecosystems, where interoperability with external systems, identity providers, analytics layers, and operational platforms may be required. The Integration & API Strategy defines the governing principles for controlled data exchange, event propagation, and ecosystem coexistence.

15.2 API-First Architectural Philosophy

The platform follows an API-centric design approach, enabling controlled interaction with system entities, operational records, automation events, and reporting datasets.

Interfaces may support:

- Secure data retrieval and submission
- Event-driven integration patterns
- Controlled external system interactions

This approach ensures extensibility without requiring architectural restructuring.

15.3 External System Coexistence

The platform is not intended to replace enterprise resource planning (ERP), manufacturing execution systems (MES), human resource systems (HRIS), or financial systems.

Instead, the platform may operate alongside such systems while focusing on:

- Operational compliance workflows
- Execution integrity
- Enforcement and traceability controls

Integration requirements may therefore vary by organizational context.

15.4 Event & Automation-Driven Interoperability

System events, including automation triggers, enforcement outcomes, and corrective workflows, may be exposed through controlled mechanisms to support interoperability scenarios.

Examples include:

- Workflow state transitions
- Compliance or failure events
- Reporting-relevant system changes

This model enables external orchestration or monitoring where required.

15.5 Data Exchange & Governance Considerations

All integrations are governed by:

- Authentication and authorization controls
- Data visibility and access policies
- Tenant isolation principles

This prevents uncontrolled cross-system data exposure.

15.6 Versioning & Compatibility Discipline

Where integration interfaces are utilized, changes follow controlled versioning principles to preserve backward compatibility and minimize operational disruption.

15.7 Architectural Outcome

This Integration & API Strategy ensures that the platform remains interoperable, extensible, and enterprise-compatible while preserving clear system boundaries and governance integrity.

16. Admin Panel

A. Dashboard - In the dashboard, it will be shown with the data's like Total Companies, Total Active Users, Recent Signups, Revenues for Subscription, Location based Signup.

B. Company Management Page

Company Creation

- Admin will create the company panel from the admin panel.

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- In the create company page, there are several pages we shall move to page by page by clicking the 'Next' button.
 - Upon entering into the creation page, the first section will be 'Company Information', it will have the fields like Company Name, Company Contact Person Name, Primary Contact Email, Primary Contact Phone number, Company Address (optional), Company Logo (optional). After entering these details click the 'Next' button to move to the next page.
 - In the next page it will be the company panel access information, by using this information only the company will do the login to their panel. In this section it will have Company's Admin Username, Company's Admin Email id, Company's Admin Phone number (optional) and click's 'Next' button.
 - In the next page it will be the company's subscription plan configuration. In this section it will have the following details.
 - Subscription Fields like Plan Name, Billing Cycle (Monthly, Annually), Start date, Renewal date, Contract Length (Monthly, Annually, Custom), Base Subscription fee, One time setup fee (optional), Price per location (optional). After given these details the next page will have the option to give the coupon based discount.
 - The Coupons & Discount section, it will be optional. If the admin needs to provide the discount on the subscription plan then the admin will generate the coupon code from the **Coupon Management Page** and provide the coupon code in this page. So the respective discount plans will be applicable on the company's billing and its cycle.
 - After providing these details click the 'Next' button.
 - In the next page it will be the Module selection section. In this section all the available modules will be shown and the admin will choose the modules which are to be granted to that company. We shall provide the checkbox or toggle button to select the modules. The available modules will be managed in the **Modules Management Page**.
 - And in the next page, it will have the option to enable the billing payment with toggle button.

- And finally there will be a Notes section, it will be used for the internal purposes. It will have the options like description box to give notes, upload option to upload any documents like contract. And once everything is added and click the 'Save' button then the invitation email will be sent to the company's admin email where the admin will click the link and complete the setup.

Management

- In the List of all registered companies will be displayed, each company will show the current status of the company's subscription (trial period, active or inactive. Also it will have filters to sort based on status, company name, email id, registered date (calendar to set the date).
- In the management page it will show all the registered companies, it will have the actions to view, edit and disable options. Admin will have the option to Edit all the details of the company, Enable/Disable the company.
- Edit Option - With the Edit option, it will navigate to the edit page where the admin will get the option to edit the company details.
- View Page In the view page, it will have separate tabs to show the details.
 - **Profile Tab**, It will show the profile details (complete profile details which is getting from the company) of the company, account created date, the last login activity (time & date), current subscription plan details. And it will have the option to Deactivate & Activate the company.
 - **Branches**, In this tab it will show companies branches based on the locations and it will have an inner page to view the details about the branches of that company (profiles details of branch).
 - **Users**, In this tab the users & its roles based on these companies will be shown, in the users will be the managers and the staff.
 - **Billing**, In this tab it will show the current subscription details as information. Admin will create the subscription plan to the company while creating the company panel and it will be activated upon the company pays from their account.

C. User Management Page

- All the users in the platform will be managed in this page, it will include all the users added under all the companies will be shown.
- In the users list admin will have the option to View, Edit, Deactivate and Password reset option.
- In this page it will have the filter options to sort the users by Company name, Status and search option by Name & Email.
- In the view page, the admin can see the complete details of that user and it will separated with the tabs,
 - **Profile**, In this tab it will show the profile information like Name, Email id, Phone number, Company name & its branch details, Status, Role.
 - **Activity Logs**, In this tab it will show all the activities done by this user as information.
 - **Login History**, In this tab all the login and logoff details will be shown.

D. Coupon Management Page

- This page is used for creating and managing the coupons for the companies subscription plans. By using this coupon code the company can get exclusive offers by the admin.
- It will have the create coupon option, it will have two types. Any one type can be used while creating the coupon.
 - **Percentage** - To give the percentage based discount, it can be generated commonly for all subscriptions plans (by selecting all) or also generated by selecting the specific subscription plans. Likewise it will have the option to select all the registered companies (by selecting all) or by selecting the specific companies (multiple selecting option).
 - **Flat Price** - To give the flat price based discount, it can be generated commonly for all subscriptions plans (by selecting all) or also generated by selecting the specific subscription plans. Likewise it will have the option to select all the registered companies (by selecting all) or by selecting the specific companies (multiple selecting option).

- For the both types it will also have the option to set the validity period (calendar option to set the date). So the coupon code will be valid at the desired date.
- **Free Trial** - The free trial period can be entered and so the actual subscription payment will begin after the trial period ends.
- **Discount Duration** - It will be the option to set type, One time (on first billing payment), Limited number of cycles (give the number of cycles the discount applies and Recurring (on every billing cycle). Any one type of discount.
- After giving the details, the coupon code will be generated (coupon code can be set by admin or system generated). The generated coupons will be managed on this page. And it will have the option to view, edit, delete and deactivate/activate options.
- Once the code is generated, it will be used by the admin while creating the company's subscription plan during the company creation process.

E. Subscription Management Page

- In the subscriptions management page, all the created subscriptions to the companies will be displayed. It will have the actions to view, edit and activate/deactivate the subscription.
- This page will have the filters to sort the subscriptions by date created (date set by calendar), active, billing cycle (monthly, annually, custom), month based (jan, feb), search option to search by company name.
- From the view action, in the inner page it will have the two tabs to show the complete details about the subscriptions of that particular company. In the 'Current Plan' tab the current subscription plan details, current due date, last payment date, transaction details and the invoice button to generate the invoice and view the invoice. And followed by next tab 'Payment History' it will show all the previous payments as a list with date, time and other required details, also for each payment transaction it will have the invoice button to generate the invoice and view it.

F. Modules Management Page

- In this page it will show all the available modules in the system, it will show the modules and action to enable/disable (toggle button) option. In this main page it will have the search filter to sort by company name, module name.
- It will have the view option, in the view inner page it will show the list of all companies who are all using this module. In the list it will show the company name, date granted, current status.
- The following are the modules available in the system.
 - Checklists & Forms
 - Inspections & Audits
 - Corrective Actions
 - Sensors & IoT
 - Labeling
 - Scheduling & Time Clock
 - Logbook/Communication
 - Document Management
 - Reporting & Analytics
 - Template Library
 - Automation & Alerts

17. Company Panel

- A. Company Dashboard** - Quick, actionable overview of company operations and compliance health.

01. UI / Sections

- Header: Company name, current plan badge, location selector (dropdown)
- KPI widgets (configurable):
 - Pending checklists (count)
 - Overdue checklists (count)
 - Open CAPAs / corrective actions (count)
 - Active alerts (count)

- Upcoming scheduled audits (list / next 3)
- Compliance score / trend (sparkline)
- Alerts feed: Latest critical alerts (temperature breach, overdue CAPA)
- Tasks widget: Assigned tasks for logged-in user (click to open)
- Recent Activity: chronological list (user, action, time)
- Quick Actions (buttons): Create Checklist | Create Audit | Create CAPA | Add User
- Reports snapshot: last-run report links, scheduled reports
- Footer: Last sync time (for devices), data freshness indicator

02. Filters / Interactions

- Time-range selector (Today / 7d / 30d / Custom)
- Drilldown: click KPI to open relevant page (e.g., click Overdue → Checklists History filtered)

B. Company Profile & Settings - Company-level configuration and preferences.

01. Page Elements

- Company information: Company name, Logo, Primary contact, Address, Time zone
- Locales & Format: Date format, Time format, Week start day
- Notification Preferences: default channels, digest settings
- Security & Access: SSO (SAML) setup status, password policy (inherited), MFA requirement
- Integrations: list of connected external systems (POS, HR, IoT), API keys (if enabled)
- Support & Onboarding status: onboarding checklist, training schedule, assigned CSM

02. Actions

- Edit company details, upload logo, configure timezone, submit support request

C. Users Management - Add, edit, assign and audit user accounts for the company.

01. Page Elements

- Toolbar: Add User (button), Bulk Import (CSV), Invite via Email (button)

- Search bar + filter chips: Role, Location, Status (Active/Inactive), Last Login range
- Users table columns: Checkbox, Avatar, Full Name, Email, Role(s), Location(s), Status, Last Login, Actions (Edit, Reset Password, Deactivate, Impersonate)
- Pagination and page size control

02. User Details Page

- Fields (editable): Full Name, Email (read-only after invite accepted), Phone, Role(s) (multi-select), Locations (multi-select), Timezone (optional), Employment ID (optional), Manager (dropdown)
- Access toggles: Active / Inactive, Requires MFA (toggle), API Access (toggle)
- Actions: Resend Invite, Reset Password, Deactivate, Audit Log (view)

03. Bulk Actions

- Assign Role, Assign Locations, Deactivate, Export selected users to CSV

D. Roles & Privileges - Define role templates and granular permissions for company-level operations.

01. Page Elements

- Role list: Role name, Description, Number assigned (count), Actions (Edit, Duplicate, Delete)
- Create Role button → opens Role Builder
- Duplicate existing role

02. Role Builder Screen

- Fields: Role Name, Description, Default landing page, Role color (optional)
- Permission groups (accordion sections by module): e.g., Checklists, Audits, CAPA, Sensors, Labeling, Scheduling, Documents, Reports, Admin
- Permissions per group (checkboxes): View, Create, Edit, Delete, Assign, Publish, Export, Configure
- Scope settings: All Locations / Assigned Locations only (radio), All Templates / Assigned Templates only
- Save as Template (checkbox) for reuse

03. Actions

- Export / Import role JSON
- Assign role to users via Users page

E. Location Management Page - Create and manage physical locations (branches/sites) for the company.

01. Page Elements

- Location list: Name, Address (city/state), Time Zone, Manager, Status (Active/Inactive), #Users, #Devices, Actions (View, Edit, Deactivate)
- Add Location button

02. Location Details Page (Tabs)

- Profile tab: Location Name, Full Address, Time Zone, Opening Hours (per day), Contact Person(s), Phone, Email, Operational Notes
- Users tab: Users assigned to this location, quick-assign controls
- Checklists tab: Templates assigned to this location (list + schedule summary)
- Sensors / Devices tab: Devices assigned, last seen, status
- Activity / Logs tab: Recent activity for this location
- Integrations tab: Connected systems (POS, ERP, IoT gateway), keys/tokens

03. Actions

- Activate / Deactivate location
- Bulk assign templates or roles to location
- Export location data

F. Checklists and Forms Module

Create, schedule and manage checklist templates and their executions.

01. Checklist Templates Page

- Columns: Template Name, Category, Industry Tag, Assigned Locations, Frequency, Status, Last Modified, Actions (Edit, Duplicate, Archive)
- Search & filters: Category, Location, Owner, Status

02. Checklist Builder Page

- Tabs: General, Fields, Schedule, Assignment, Compliance Settings, Preview

- Fields in General: Template Name, Description, Category (dropdown), Industry Tag (dropdown), Owner (user), Version notes
- Fields tab: drag-and-drop area to add Field Types (Text, Number, Yes/No, Pass/Fail, Dropdown, Multi-choice, Photo, Signature, Temperature, Barcode)
 - Per-field properties: Label, Help Text, Placeholder, Required (toggle), Scoring Weight, Conditional Logic (define one or more show/hide rules), Attach SOP (link), Photo Required (toggle)
- Schedule tab: Recurrence (None, Daily, Weekly, Monthly, Custom), Time, Start/End date, Grace period, Reminders
- Assignment tab: Assign to Role / User / Location, Allow ad-hoc runs (toggle)
- Compliance Settings: Auto-create CAPA on fail (toggle), Require signature, Require photo for failed items
- Preview tab: Desktop & Mobile preview, Test run

03. Scheduled Instances / Running Checklists

- View upcoming, in-progress, completed, missed instances
- Actions: Reassign, Cancel, Re-run, Add note

04. Checklist Submissions

- List of instances with columns: Template, Location, Completed By, Date/Time, Status, Score, Actions (View, Export, Create CAPA)
- Filters: Date range, Location, User, Template, Status

05. Checklist Details

- Full answers, photos (lightbox), timestamps, geolocation, signature, linked CAPAs, comments, audit trail

06. Actions

- Publish / Unpublish template
- Duplicate template
- Archive template
- Export instances (CSV / PDF)
- Bulk assign templates to locations

G. Inspections & Audits Module (Available when enabled)

The Inspections & Audits Module allows Company Admins and designated inspectors to create, schedule, perform, and review structured audits. These audits typically contain multiple sections, weighted scoring, evidence requirements (photos, comments), and multi-level sign-offs. The module supports compliance audits, operational inspections, safety assessments, hygiene audits, facility inspections, and more.

01. Audit Templates

Page Contents

- Template Name
- Category (Safety / Hygiene / Operations / Compliance / Custom)
- Number of Sections
- Assigned Locations (multi-location support)
- Last Modified (Date & Time)
- Status: Active / Inactive / Draft
- Actions: View, Edit, Duplicate, Archive/Restore

Filters & Searches

- Search by Template Name
- Filter by Category
- Filter by Assigned Locations
- Filter by Status (Active / Inactive / Draft)
- Filter by Created By

Button

- Create new Audit Template

02. Audit Templates Builder(create/edit)

2.1) General Settings Page

Fields

- Audit Template Name

- Description
- Category (dropdown list)
- Industry Tag (optional)
- Assigned Locations (multi-select)
- Assigned Roles (who can perform the audit)
- Template Owner
- Enable Scoring (toggle) - If ON → choose scoring type
- Scoring Type - Pass/Fail, Weighted Scoring (0–100), Section-based Scoring

Sign-Off Workflow

- Enable Multi-Level Sign-Off (toggle) and If enabled, sign-off order must be followed - Level 1: Inspector, Level 2: Supervisor, Level 3: Quality Manager

2.2) Sections & Questions Tab - Audits consist of sections; each section contains questions.

Section-Level Actions

- Add Section
- Edit Section Name
- Add Section Description
- Assign Section Weight (if scoring enabled)
- Reorder Sections (drag & drop)
- Delete Section (with confirmation)

Add Question (Field Types) - Each section can include:

- Yes / No
- Yes / No / N/A
- Pass / Fail
- Multiple Choice (single-select)
- Multiple Choice (multi-select)
- Numeric Input
- Text Input (comments)
- Photo Capture Required

- File Upload
- Signature Field

Per Question Options

- Required (toggle)
- Add Helper Text (instruction field)
- Add Placeholder
- Add Scoring Value or Weight
- Conditional Logic: Show this question only if previous answer = Fail/Yes/No
- Photo Required on Fail (toggle)
- Comment Required on Fail (toggle)
- Attach Reference Document (link to SOP/PDF)

2.3) Scheduling Tab - If the audit should occur regularly, settings include:

Scheduling Options

- Schedule Type: None (manual audits only), One-Time, Daily, Weekly, Monthly, Custom (multiple selected days)
- Start Date
- End Date
- Time of Day
- Reminder Notification (minutes/hours before)
- Grace Period for Completion

Inspector Assignment

- Assign to: Specific Users, Roles, Specific Locations
- And if Multiple auditors are assigned: Choice: "Any auditor may complete" or All auditors must complete their own sections"

2.4) Assignment Tab - Configure who is responsible for performing the audit.

Fields

- Assigned Roles
- Assigned Users
- Assigned Locations
- Restrict Execution to Assigned Auditors Only (toggle)

- Enable Collaborative Audits (multiple inspectors contribute to same audit)

2.5) Compliance & Workflow Settings

- Require photo for any failed response (toggle)
- Automatically create Corrective Action for failures (toggle)
- Force Supervisor Review before final submission (toggle)
- Enable digital signatures (toggle)

2.6) Actions

- Save as Draft, Publish Template, Duplicate Template

03. Audit Scheduler - Manage all scheduled audits & recurring inspection tasks.

Page Contents

- Audit Name
- Template
- Assigned Location(s)
- Assigned Inspector(s)
- Frequency (Daily/Weekly/Monthly/Custom)
- Next Scheduled Date & Time
- Status (Active / Paused / Expired)

Actions

- Create New Schedule
- Edit Schedule
- Pause / Resume Schedule
- Delete Schedule
- Run Now (start audit immediately)

Filters

- Filter by Audit Template
- Filter by Location
- Filter by Assigned User
- Filter by Frequency

04. Active Inspections - Show audits currently in progress.

Page Contents

- Audit Name
- Template Name
- Location
- Assigned Inspector(s)
- Start Time
- Due Time
- Progress Bar (% completed)
- Status: In Progress, Pending Supervisor Review, Pending Manager Approval

Actions

- Continue Audit
- Reassign Inspector
- Add Notes
- Cancel Audit

05. Performing an Audit

Screen Layout

- Sections listed with progress indicator
- Expand or collapse each section
- Questions displayed one at a time or in list form
- Photo capture button
- Add comments button
- Attach files
- Fail-based prompts for mandatory evidence

Features

- Auto-save responses
- Offline mode (mobile)
- Add general comments
- “Mark Section Complete” button
- “Submit Audit for Review” (if approval enabled)

Signature

- Inspector must sign before submission (if required)

06. Audit Sign-Off & Workflow

Level 1: Inspector Submission

- Inspector completes audit
- Submits for approval (if enabled)

Level 2: Supervisor Review

- Supervisor reviews responses
- Confirms evidence
- Approves or rejects audit
- Rejection requires comment and returns audit to inspector

Level 3: Manager / Quality Approval

- Final approval before audit is closed
- Manager may override scores or add final comments

07. Completed Audits - Show all completed & approved audits

Columns

- Audit Name
- Location
- Completed By
- Completed On
- Final Score
- Status
- Linked CAPAs
- Actions (View Report, Export PDF)

Filters

- Date range
- Location
- Template
- Score Range
- Inspector
- Status

08. Audit Reports(detail view)

Report Sections

- Audit Overview
 - Template Name
 - Location
 - Date/Time
 - Inspector(s)
 - Final Scoring Summary
- Section Breakdown
 - Section Scores
 - Section Notes
- Question Level Details
 - Question
 - Response
 - Photos (expandable viewer)
 - Comments
 - Fail indicators
- Sign-off Log (Inspector → Supervisor → Manager)
- Linked Corrective Actions
- Export Buttons - Generate PDF, Export CSV

09. Audit Findings(non-conformances)

Contents

- Finding Summary
- Severity
- Origin Audit
- Location
- Assigned To
- Status (Open / In Progress / Closed)

Actions

- Convert Finding into CAPA
- Assign User
- Add Evidence
- Change Status

Filters

- Severity
- Location
- Date
- Assigned User
- Status

H. Corrective ActionsModule (Available when enabled)

Track non-conformances from detection to closure with evidence and SLA.

01. Corrective Actions – Overview Page

Page Contents

- Summary Widgets - Total Open CAPAs, Overdue CAPAs, CAPAs Assigned to Me, CAPAs Completed This Month, CAPA Closure Rate (%)
- CAPA Status Chart - Open, In Progress, Awaiting Verification, Closed
- Severity Breakdown Chart - Low, Medium, High, Critical
- Quick Filters - Location, Severity, Status
- Latest CAPA Activity Feeds - Created, assigned, updated, closed events

02. Open CAPAs Page

Table Columns

- CAPA ID (auto-generated)
- CAPA Title
- Origin (Checklist / Audit / Sensor / Manual)
- Location
- Severity (Low / Medium / High / Critical)
- Assigned To (User or Role)
- Created Date
- Due Date
- Progress Status (Open / In Progress / Awaiting Verification)
- Actions (View, Reassign, Change Status)

Filters

- Severity
- Origin Type

- Assigned User
- Location
- Due Date range
- Status

Bulk Actions

- Assign Selected
- Change Status
- Export Selected

03. Assigned to Me

Contents

- CAPAs filtered automatically by assignee
- Quick-highlight overdue CAPAs assigned to the user
- Priority badge (High/Critical)

Actions for each CAPA

- Update Progress
- Add Evidence (photo/file upload)
- Add Comment
- Submit for Verification
- Request Extension

04. Overdue CAPAs

Table Columns

- CAPA ID
- CAPA Title
- Origin
- Severity
- Assigned User
- Due Date
- Days Overdue
- Actions (Escalate, Reassign)

Actions

- Escalate to Supervisor / Manager

- Notify the Assigned User
- Change Due Date (Company Admin only)

05. Completed CAPAs

Table Columns

- CAPA ID
- Title
- Origin
- Location
- Severity
- Completed By
- Completed Date
- Verification Status (Verified / Reopened)
- Attachment/Evidence Indicator
- Actions (View Report, Export)

Filters

- Date Range
- Verification Status
- Severity
- Location
- Origin Type

06. CAPA Details Page

CAPA Header - Date Range, Verification Status, Severity, Location, Origin Type

6.1) Summary Section

- CAPA Title
- CAPA Description
- Origin Details
- Created By (Name, Date, Time)
- Assigned To (user or role)

6.2) Root Cause Analysis Section - Allows Company Admin or Managers to input structured RCA.

Fields

- 5 Whys Analysis fields
- Root Cause Description
- Contributing Factors Field
- Attach Supporting Documents

6.3) Actions/Tasks Section - Each CAPA may contain one or more tasks.

Fields Per Task

- Task Title
- Description
- Assigned User
- Due Date
- Attachments (photos/files)
- Status (Not Started / In Progress / Completed)

Actions

- Add New Task
- Assign/Reassign User
- Set Priority
- Mark Complete
- Add Evidence

6.4) Evidence Section

- Photos (gallery)
- File uploads
- Notes/comments
- Timestamp and user details

6.5) Activity Log Section - A chronological audit trail including

- CAPA created
- Status changes
- Reassignments
- Evidence uploads
- Comments
- Completion
- Verification

6.6) Verification Section

Once tasks are completed, For Company Admin or Manager:

- Approve Verification → CAPA CLOSED
- Reject Verification → CAPA REOPENED
- Add Verification Notes
- Upload verification evidence
- Verification is mandatory before the CAPA can be closed.

07. CAPA Creation Page - It can be created in four ways.

7.1) Manual Creation - Company Admin can click Create CAPA, enter all fields manually.

Fields: Title, Description, Location, Severity, Assign To, Due Date, Attach Evidence.

7.2) From Checklists Failure - If Module 1 was configured with: "Auto-generate CAPA on failure". Then a failed checklist item automatically generates a CAPA.

Prefiled Fields: Origin: Checklist, Failed question, Photo evidence, Assigned User based on rules.

7.3) From Audit Findings - Audit reviewer can click: "Create CAPA from Finding"

Prefiled Fields: Audit Reference, Section, Non-conformant question, Photos, Severity.

7.4) From Sensor Alert - If a temperature sensor breaches threshold: "Auto-create CAPA" (if rule enabled)

Origin is marked as: Sensor ID, Timestamp, Temperature value

I. Sensors & IoT Module (Available when enabled)

01. Sensors List Page - Display all sensors installed across locations.

Table Columns

- Sensor Name
- Sensor ID / Serial Number

- Sensor Type (Temperature / Humidity / Door / Multi-sensor)
- Location
- Status: Online, Offline, Battery Low, Fault
- Last Reading
- Last Seen Time
- Assigned Equipment (optional)
- Actions (View, Edit, Move, Deactivate)

Filters & Search

- Search by Sensor Name or ID
- Filter by Sensor Type
- Filter by Location
- Filter by Status (Online/Offline/Fault)
- Filter by "Threshold breached"

Bulk Action

- Move to Different Location
- Refresh Status
- Export List (CSV)

Buttons

- Add Sensor & Add Multiple Sensors

02. Add Sensor Page (Registration)

Step 1: Sensor Information

- Sensor Name
- Serial Number / QR Scan
- Sensor Type (dropdown)
- Manufacturer (optional)
- Firmware Version (auto-read if connected)

Step 2: Sensor Location

- Select Location (dropdown)
- Assign Zone or Area (fridge, freezer, warehouse rack, etc.)
- Assign Equipment (optional)

Step 3: Threshold Configuration (based on the sensor type)

Temperature Sensor:

- Minimum Allowed Temperature
- Maximum Allowed Temperature
- Alert if temperature outside range for: Immediate, After X minutes (delay to avoid false alarms)

Humidity Sensor:

- Min and max threshold values

Door Sensor:

- Alert if door open longer than X seconds/minutes

Step 4: Alert Routing (Company Admin defines who gets alerted)

- Notify Users (multi-select)
- Notify Roles (e.g., Managers)
- Notify Locations
- Notification Channels: App Push Notification, Email

Step 5: Final Configuration

- Sampling interval (every 1 / 5 / 10 minutes)
- Battery level alerts (toggle)
- Auto-create CAPA on alert (toggle)

Action

- Save & Register Sensor
- Test Sensor Reading
- Cancel

03. Sensor Details Page - Display complete information and real-time status of a single sensor.

3.1) Overview

- Sensor Name
- Serial Number
- Sensor Type
- Location
- Zone / Equipment
- Current Status (color-coded)

- Last Seen Time
- Current Temperature/Humidity Value
- Battery Level

3.2) Real-Time Reading Graph

- Live reading graph (last 24 hours, auto-refresh)
- Threshold lines (min & max)
- Hover to see actual values

3.3) Threshold Settings

Fields

- Min Temperature
- Max Temperature
- Delay Before Alert
- Alert Recurrence Behavior (notify again after X minutes)

Actions

- Edit Thresholds
- Save Changes
- Reset to Defaults

3.4) Alert Routing - Shows a list of users/roles who will receive alerts.

Actions

- Add User
- Remove User
- Change Notification Channels

3.5) Calibration Info

- Last Calibration Date
- Calibrated By
- Calibration Certificate Upload
- Next Calibration Due Date

Actions

- Upload Certificate
- Mark Calibration Completed

3.6) Activity Log

- Sensor added
- Threshold changed
- Alerts triggered
- Alerts acknowledged
- Sensor offline/online events
- Battery alerts
- All entries show: User, Timestamp, Description

04. Live Monitoring Dashboard

4.1) Dashboard Widgets

- Total Sensors
- Sensors Online
- Sensors Offline
- Active Alerts
- Average Temperature (by location)

4.2) Live Sensor Grid (column)

- Sensor Name
- Location
- Last Reading
- Threshold Status: Normal, Above Max, Below Min, Alert Active
- Status Badge (Online/Offline)

4.3) Actions

- Click any sensor to open details
- Group sensors by location
- Display temperature graph overlay

05. Alerts Inbox

5.1) Alert Table Column

- Alert ID
- Sensor Name
- Location
- Alert Type (Temperature High/Low, Door Open, Battery Low)
- Reading (e.g., 12°C vs allowed 5°C)

- Time Triggered
- Status: New, Acknowledged, CAPA Created, Resolved
- Actions: View Alert, Acknowledge, Create CAPA, Add Notes

5.2) Filters

- Alert Type
- Status
- Location
- Sensor
- Date range

06. Alerts Details Page

6.1) Alert Overview

- Sensor Name
- Type of alert
- Actual value
- Threshold violated
- Time detected
- Time acknowledged (if applicable)

6.2) Sections

- Alert Readings Chart - Graph of readings before and after the alert
- Evidence & Notes - Photo from staff (e.g., showing freezer door closed), Comment section for managers
- Actions Available
 - Acknowledge alert
 - Assign CAPA to responsible staff
 - Add notes or images
 - Mark alert as resolved
- Auto CAPA Link - If enabled in settings display
 - CAPA ID
 - CAPA status
 - View CAPA button

07. Sensors Log Page/History

7.1) Features

- Graph by: Hour, Day, Week, Month
- Export readings to: CSV, PDF

7.2) Table Logs

- Timestamp
- Reading value
- Status (Normal/Above/Below)

7.3) Filters

- Date range
- Location
- Sensor type

08. Calibration & Maintenance App

8.1) Contents

- Calibration schedule
- List of sensors requiring calibration
- Past calibration history
- Calibration certificate uploads
- Technician contact details

8.2) Actions

- Log calibration
- Upload certificate
- Mark sensor as “Out of Service”

09. Sensor Settings Page (Company Admin Only)

General Settings

- Default alert delay
- Default alert recipients
- Notification frequency
- Auto-CAPA rules

Device Management Options

- Data retention settings
- Enforce offline alert if sensor is offline > X minutes

- Auto-disable inactive sensors
- Automatic recalibration reminders

J. Labeling Module (Available when enabled)

Manage print-ready labels for products, batches, expiry, allergens.

01. Label Templates Page

Table Columns

- Template Name
- Label Type (Expiry / Batch / Product / Allergen / Custom)
- Dimensions (e.g., 2×3 inches)
- Assigned Locations
- Last Modified Date
- Status (Active / Inactive)
- Actions (Edit, Duplicate, Print, Archive)

Filters

- Label Type, Location, Status, Created By

Buttons

- Create New Label Template
- Import Label Template
- Export Templates

02. Create/Edit Label Template (Label Designer)

2.1) Template Settings

- Template Name
- Label Type (dropdown)
- Label Size: Width (mm/inch), Height (mm/inch)
- Orientation: Portrait / Landscape
- Default Printer (optional)
- Assigned Locations

2.2) Canvas Area (Drag & Drop) - Admins can drag and place elements onto the label canvas.

Available Elements

-
- Text
 - Dynamic Variables:
 - Product Name
 - Prepared Date
 - Expiry Date / Use-By Date
 - Batch Number
 - Employee Name
 - Allergen List
 - Location Name
 - Custom Fields
 - Shapes (rectangles, lines)
 - Barcode (Code128, EAN)
 - QR Code
 - Image (e.g., company logo)
 - Icons (allergen symbols)

Element Properties (Right Panel) - For each element, Company Admin can configure.

- Font size
- Font style (bold, italic)
- Text alignment
- Field binding (for dynamic variables)
- Border properties
- Background color
- Position coordinates (X, Y)
- Size (Width, Height)
- Rotation
- Condition visibility (e.g., only show allergen icons if flagged)

2.3) Variable Mapping Panel

Standard Variable

- Product
- Category

- Prepared By (employee name)
- Prepared Date (auto-generated)
- Expiry Date (calculate using shelf-life settings)
- Allergen Flags
- Batch No (manual input or auto-increment)

Custom Variable (company admin may create)

- Numerical fields
- Static dropdowns
- Text fields
- Calculated fields (e.g., expiry = prepared date + X hours)

2.4) Shelf-Life Rules - For food/hospitality or pharma processes.

Fields

- Shelf-life duration (hours/days)
- Auto-calculate expiry based on:
- Preparation time
- Production batch time
- Override allowed (toggle)

2.5) Preview Panel - Shows real label preview with sample variable data.

Actions

- Preview Actual Size
- Export Preview (PDF)
- Test Print

2.6) Template Actions

- Save Template
- Save as Draft
- Publish Template
- Duplicate Template
- Archive Template

03. Print Labels Page - Company admin or authorized person will use.

3.1) Template Selection

- Dropdown: Select Label Template. Once selected → displays required input fields.

3.2) Data Entry Form - Based on the variable defined in templates. Say,

- Product Name (textbox)
- Batch No. (textbox or auto-generated)
- Prepared Date (auto-filled, editable or locked)
- Shelf-Life → Auto-calculate expiry
- Allergens (multi-select)
- Quantity (number of label copies)

3.3) Printer Selection (dropdown selection)

- Default Printer
- Available Network Printers
- Bluetooth Printers (mobile)

3.4) Preview & Print (buttons)

- Preview Label (PDF)
- Print Now
- Download PDF

04. Print History

4.1) Table Columns

- Print ID
- Template Name
- Printed By
- Location
- Date/Time
- Variables used (Product, Prepared Date, Expiry, Batch No., etc.)
- Number of Copies
- Printer Used
- Actions: View Details, Reprint, Export PDF

4.2) Filters

- Date range
- Template

- Printed By
- Location

05. Printer Management

5.1) Printer List Columns

- Printer Name
- Printer Type (Zebra, Bluetooth, Thermal)
- IP Address / Connection Type
- Assigned Location
- Status (Online / Offline)
- Default Printer (toggle)
- Actions (Edit, Test Print, Remove)

5.2) Add Printer Fields

- Printer Name
- Printer Type
- Connection Method: IP, USB (desktop only), Bluetooth (mobile)
- IP Address / Pairing Code
- Assign to Location

5.3) Actions

- Add Printer
- Test Connection
- Set Default

06. Label Variables & Data Source

- Standard Variables - Prepared Date, Expiry Date, Employee Name, Location Name
- Custom Variables - Text, Number, Dropdown, Boolean, Auto-generated counter (e.g., BATCH-001)
- Actions: Create Variable, Edit Variable, Delete Variable

07. Label Settings Page

7.1) Settings Options

- Default Label Size
- Default Expiry Rule: Prepared Date + X hours, Prepared Date + X days

- Naming Convention for Auto-generated batch numbers
- Reprint Permissions (who can reprint labels)
- Auto-log all printed labels (toggle)
- Required fields before printing (enforce data completeness)

7.2) Advanced Options

- Enable QR Codes to link to internal checklist/audit documents
- Enable multi-language labels
- Allow staff to print or require manager approval

K. Scheduling & Time Clock Module (Available when enabled)

Manage staff rosters, track attendance and time-off.

01. Schedule Calendar

View Options: Week View (default), Day View, Month View, Location Filter (single or multi-location), Role filter (e.g., cashier, cook, supervisor), **Shift Display (each shift block shows):** Employee Name, Role, Shift Time (Start-End), Location, Shift Status: Draft / Published / Updated, **Actions:** Create Shift, Edit Shift, Move Shift, Copy Shift, Delete Shift, Assign/Unassign Employee

1.1) Shift Creation Model

Fields

- Select Employee
- Role
- Location
- Start Time
- End Time
- Break Duration
- Notes (optional)
- Tags (e.g., Training Shift, Extra Shift)

Advanced Options

- Mark as Overtime Shift
- Lock shift (cannot be modified by managers)

- Allow Employee to Swap Shift
- Allow Employee to Drop Shift

Actions

- Save, Save & Create Another, Cancel

1.2) Publishing Schedules - When shifts for the week/month are ready.

Button: Publish Schedule

Upon Publishing:

- Assigned employees receive:
 - Push notification
 - Email (optional)
- Schedule changes after publication require:
 - "Republish changes"
 - Notification to affected users

Versioning:

- Schedule Version 1
- Updated Schedule Version 2
- Change log visible to Company Admin

02. Shift Template Library

2.1) Table Columns

- Template Name
- Role
- Default Start Time
- Default End Time
- Default Break Duration
- Assigned Locations
- Actions (Edit, Duplicate, Delete)

Create Shift Template Fields

- Template Name
- Role
- Default Times
- Default Break

-
- Color Coding (optional)

Usage: Admins can drag a shift template from the sidebar onto the calendar to create a shift instantly.

03. Time Clock Dashboard

3.1) Dashboard Elements

- Total Employees Clocked In
- Late Clock-ins
- Missed Clock-ins
- Currently Clocked Out
- Overtime Alerts

3.2) Live Activity Feed (Ex)

- "John D. clocked in at 08:02 AM (2 minutes late)"
- "Maria L. ended shift early (Clock-out 4:43 PM)"

3.3) Features

- Location filter
- Real-time updates every X seconds

3.4) Actions

- View employee time card
- Manual Clock-In/Clock-Out override (Admin only)
- Flag time entry for review

04. Time Entries Page

4.1) Table Columns

- Employee Name
- Role
- Location
- Date
- Clock-In Time
- Clock-Out Time
- Total Hours
- Break Time

- Overtime Hours
- Verification Status
- Flags (Late / Missed / Irregular)

Actions (Edit, Approve, Delete)

- Filters
- Employee
- Location
- Date range
- Irregular entries
- Overtime

Manual Edits (Admin Only)

- Add missing time entries
- Adjust incorrect times
- Add comments
- Approve or reject flagged events

All changes appear in the Audit Log.

05. Time-Off Requests Page - If enabled staff may request time-off

5.1) Table Column

- Request ID
- Employee
- Time-Off Type (Vacation, Sick, Unpaid, etc.)
- Start Date
- End Date
- Submitted Date
- Status (Pending / Approved / Rejected)
- Actions (Approve, Reject, Add Notes)

5.2) Actions

- Bulk approve or reject requests
- View employee remaining balance (if tracking PTO)
- Add manual time-off entry

06. Availability Page - Company Admin can set or review employee availability.

Availability is used to validate schedule conflicts.

6.1) Table Columns

- Employee Name
- Weekly Availability (Mon–Sun blocks)
- Preferred Locations
- Restrictions (Cannot work nights, etc.)

6.2) Actions

- Edit employee availability
- Add restrictions
- Approve employee-submitted availability changes

07. Kiosk Management Page

7.1) Supported Methods

- PIN-based login
- QR badge scanning
- Face recognition (optional)

7.2) Kiosk List Columns

- Kiosk Name
- Device Type
- Location
- Status
- Allowed Users
- Actions (Configure, Reset PINs, Deactivate)

Configure Kiosk (fields):

- Clock-in methods allowed
- Geofencing radius (if mobile kiosk)
- Idle logout time
- Force photo capture on clock-in/out (optional)

08. Payroll Export Page

8.1) Export Formats - CSV, Excel

8.2) Export Options

- Select date range
- Select locations
- Include overtime
- Include flagged entries
- Include notes

09. Scheduling & Time Clock Settings Page

9.1) Scheduling Rules

- Allow shift swapping (toggle)
- Allow shift dropping (toggle)
- Minimum rest time between shifts
- Maximum daily/weekly hours
- Warn if overlapping shifts
- Required roles per shift (e.g., at least 1 supervisor & 1 line cook)

9.2) Time Clock Rules

- Geofencing (radius in meters)
- Photo capture required on: Clock-in & Clock-out
- PIN length
- Auto-clock-out after X hours
- Break enforcement:
- Required break every X hours
- Auto-deduct break duration

9.3) Overtime Rules

- Daily overtime threshold
- Weekly overtime threshold
- Approval required before overtime
- Notify managers when overtime is triggered

9.4) Notifications

- Notify staff of new schedules
- Notify staff of shift changes
- Notify admin of late clock-ins
- Notify admin of missed shifts

L. Logbook/Communication Module (Available when enabled)

Central place for shift handovers, incident notes, announcements and acknowledgements.

01. Logbook Feed - Display all log book entries

1.1) Feed Structure - Entries appear in chronological order with the following details visible:

- Entry Type: Shift Note, Incident, Announcement, Handover Note, Task Follow-up
- Title
- Preview of Content
- Location
- Created By (user)
- Date & Time
- Attachment Indicator (photos/files)
- Acknowledgement Status (if required)
- Tags (e.g., "Urgent", "Equipment", "Cleaning")

1.2) Filters & Search

- Search (keyword)
- Filter by: Location, Date range, Entry Type, Tags, Status (Acknowledged / Pending)

1.3) Actions

- View Entry
- Edit Entry
- Pin Entry
- Unpin Entry
- Archive Entry

02. Create New Log Entry

2.1) Entry Types

- Shift Note
- Announcement

- Incident Report
- Task Handover
- General Note

2.2) Fields

- Entry Title
- Entry Type (dropdown)
- Location (dropdown)
- Description / Content (rich text)
- Tags (multi-select): safety, operations, equipment, hygiene, urgent
- Attachments:
 - Photos
 - Videos (optional)
 - Documents (PDF, DOCX)
- People to Notify (optional): Users, Roles
- Require Acknowledgement? (toggle) - If yes → select required users or roles
- Visibility Options:
 - Visible to all roles
 - Visible to managers only
 - Visible to specific roles
- Follow-up Required (toggle)
- Due Date (if follow-up is enabled)

2.3) Actions

- Save Entry
- Publish Entry
- Save as Draft

03. Log Entry Details Page

3.1) Entry Overview

- Title
- Type
- Created By

- Created Date & Time
- Location
- Tags
- Follow-up Due Date (if applicable)

3.2) Entry Content

- Full description
- Photos
- Attachments
- Linked CAPA (if incident required corrective action)
- Linked tasks (handover tasks or follow-up tasks)

3.3) Acknowledgement Status

If “Require Acknowledgement” is enabled:

- List of users who acknowledged
- List of users who have not yet acknowledged
- Timestamp of acknowledgements

Admin Actions:

- Send reminder
- Add additional required users

3.4) Common Sections

- Authorized users may add comments. Each comment shows: User, Timestamp, Edited status

3.5) Activity Log - Tracking

- Entry created
- Entry edited
- Entry acknowledged
- Attachments added
- Entry pinned/unpinned
- Entry archived
- Updates to required acknowledgements

04. Acknowledgements Dashboard - Allow company admin to view entries.

4.1) Dashboard Widgets

- Total Entries Requiring Acknowledgment
- Acknowledged by All
- Pending Acknowledgements
- Overdue Acknowledgements

4.2) List/Table Columns

- Entry Title
- Location
- Required Roles/Users
- Acknowledgement Status
- Due Date (if set)
- Actions: View Entry, Send Reminder

4.3) Filters

- Entry Type
- Location
- Date
- Status

05. Incident Reports Page

5.1) Columns

- Incident ID
- Title
- Severity (Low/Medium/High/Critical)
- Location
- Reported By
- Date & Time
- Status (Open/Under Review/Closed)
- Actions:
 - View
 - Convert to CAPA
 - Assign Reviewer

5.2) Fields in Incident Report (Extended) - When creating an incident log entry, if "Incident Report" type is selected:

Additional fields appear:

- Severity Level
- People Involved
- Equipment Involved
- Immediate Action Taken
- Injuries (Yes/No)
- Require CAPA? (toggle)

If CAPA is required:

- Auto-generate CAPA (optional)
- Link CAPA ID

06. Logbook Categories / Tags Management

6.1) Table Columns

- Tag Name
- Color (optional)
- Usage Count
- Status
- Actions (Edit, Delete, Disable)

6.2) Actions

- Add new tag
- Merge tags (Admin only)
- Assign tags to departments or locations

07. Logbook Settings (Company Admin Only)

7.1) General Settings

- Allow staff to create log entries (toggle)
- Allow managers to edit staff entries (toggle)
- Enable incident severity levels
- Default acknowledgment settings
- Allow attachments (toggle)
- Maximum number of attachments

7.2) Acknowledgement Rules

- Require acknowledgment for Announcement type entries

- Reminders frequency (daily/weekly)
- Automated overdue alerts

7.3) Follow-Up Rules

- Default due date for follow-up items
- Require manager approval for closing follow-ups

7.4) Sharing & Visibility Rules

- Default visibility by role
- Restrict incident reports to managers only (toggle)

M. Document Management Module (Available when enabled)

Store SOPs, policies; manage versions and acknowledgements.

01. Document Library

1.1) Table Columns

- Document Title
- Document Type (SOP / Policy / Training Manual / etc.)
- Assigned Locations
- Last Modified Date
- Version Number
- Required Acknowledgement (Yes/No)
- Status (Published / Draft / Archived)
- Actions (View, Edit, Replace, Publish/Unpublish, Archive)

1.2) Filters & Search

- Search by document name
- Filter by document type
- Filter by assigned location
- Filter by status
- Filter by acknowledgement requirement

1.3) Actions

- Upload New Document
- Create Folder
- Move Document

- Export List (CSV)

1.4) Structure - Documents can be organized into:

- Folders
- Subfolders
- Tags

02. Upload New Document (Fields)

- Document Title
 - Document Type (dropdown)
 - SOP
 - Policy
 - Manual
 - Form
 - Safety Regulation
 - Custom
- Upload File (PDF, DOCX, XLSX, PPT, JPG/PNG allowed)
- Description (rich text)
- Assigned Locations (multi-select)
- Visibility Options
 - All Users
 - Managers Only
 - Specific Roles
- Require Acknowledgement? (toggle)
 - If enabled → Choose required roles/users
- Review Frequency (optional)
 - 6 months
 - 12 months
 - Custom
- Effective Date (optional)
- Tags (multi-select)

03. Document Details Page

3.1) Document Overview

-
- Title
 - Description
 - Type
 - Created By
 - Created Date
 - Last Modified
 - File Preview Thumbnail
 - Assigned Locations
 - Visibility Rules
 - Required Acknowledgement (Yes/No)
 - Version Number

3.2) Document Preview

- Inline PDF viewer (or embedded preview for other formats)
- Full screen mode
- Download button
- Print button

3.2) Acknowledgement Status (Only visible if acknowledgment is required)

Shows:

- Number of users required to acknowledge
- Number acknowledged
- Number pending
- List of users with:
 - Name
 - Role
 - Location
- Acknowledgement timestamp
- Send Reminder button

3.3) Comments & Notes (Admin Only) Fields:

- Add comment
- Pin comment
- View comment history

3.4) Document Actions

- Replace Document (upload new version)
- Edit Details
- Manage Acknowledgements
- Move to Another Folder
- Archive Document
- Delete Document (only if no active version exists)

04. Version History Page - Tracks document changes over time.

4.1) Version Log Columns

- Version Number
- Uploaded By
- Upload Date
- Change Summary
- Download Old Version
- Restore This Version (Admin Only)

4.2) Actions

- Add version notes
- Compare versions (thumbnails)

4.3) Version Control Rules

- Only one active version at a time.
- Old versions remain readable but not assignable.
- Restoring an old version creates a new version entry.

05. Acknowledgement Dashboard

5.1) Dashboard Widgets

- Total Required Acknowledgements
- Completed Acknowledgements
- Pending Acknowledgements
- Overdue Acknowledgements (based on effective date)

5.2) Table Columns

- Document Title
- User Name

- Role
- Location
- Due Date
- Status (Acknowledged / Pending)
- Acknowledged On
- Actions: Send Reminder

5.3) Filters

- Document
- User
- Role
- Location
- Status

5.4) Bulk Actions

- Send Reminder to Selected Users
- Export Acknowledgement Report (PDF/CSV)

06. Folder Management Page

6.1) Functions

- Create Folder
- Create Subfolder
- Rename Folder
- Delete Folder (only if empty)
- Move Documents Between Folders

6.2) Folder Permissions - Set folder visibility by role or location

07. Document Settings Page (company admin)

7.1) General Settings

- Allowed file types
- Maximum file size
- Default document type
- Default visibility (all users/managers only)
- Require acknowledgement by default (toggle)

7.2) Acknowledgement Settings

- Send auto-reminders every X days
- Mark unacknowledged users as overdue after X days
- Require acknowledgement on every new version (toggle)

7.3) Review Settings

- Auto-notify admin when a document review date approaches
- Create task for reviewer
- Escalate if overdue

7.4) Security Settings

- Enable watermarking on internal documents
- Prevent downloading except for authorized roles
- Restrict printing permissions

N. Reporting & Analytics Module (Available when enabled)

The Reporting & Analytics module provides companies with insights into operational performance, compliance, task completion, employee activity, sensor status, and more.

01. Reports Dashboard

1.1) Dashboard Widgets

- Completion Rate for Checklists
- Audit Score Trends
- Open vs Closed CAPAs
- Overdue Actions
- Sensor Alert Count
- On-Time vs Delayed Schedule Compliance
- Employee Activity Summary
- Document Acknowledgement Completion Rate

1.2) Charts

- Line Chart: Checklist Completion Trend
- Pie Chart: CAPA Status Distribution
- Bar Chart: Audit Average Score by Location
- Heat Map: Sensor Alerts by Time of Day
- Column Chart: Top Non-Conformance Categories

1.3) Filters

- Location
- Date Range
- Role (optional)
- Module Type

02. Checklists Reports

2.1) Checklist Completion Summary

- Columns:
 - Checklist Name
 - Location
 - Scheduled Count
 - Completed Count
 - Missed Count
 - Completion Rate (%)
 - Average Score (if scoring enabled)
- Filters:
 - Date range
 - Location
 - Checklist template
 - User

2.2) Missed / Overdue Checklists

- Columns:
 - Checklist
 - Scheduled Date
 - Assigned User
 - Status (Missed / Overdue)
 - Reason (optional)
- Actions:
 - Export
 - Notify User
 - Reassign Checklist

2.3) Checklist Score Summary

- Table Columns
 - Checklist Template
 - Location
 - Date Completed
 - Completed By
 - Total Score (%)
 - Pass/Fail (if applicable)
 - Number of Failed Questions
 - Time Taken to Complete
 - Linked CAPA Count
- Charts
 - Score Trend Line Chart (Over time)
 - Heat Map of average scores per location
 - Top 10 lowest scoring checklists
- Filters
 - Date range
 - Template
 - Location
 - User
 - Score range
 - Pass/Fail
- Actions
 - View checklist details
 - Export results (CSV/PDF)
 - Create CAPA (if applicable)

2.4) Checklist Question Analysis

- Columns:
 - Question
 - Checklist Template
 - Fail Rate (%)

-
- Total Responses
 - Locations with Most Failures
 - Chart:
 - Bar chart of failures per question
 - Additional Metric:
 - Fail Ratio (%) across all submissions
 - Heat Map by location
 - Ability to drill down into individual responses

2.5) Checklist Activity by User

- Columns:
 - User Name
 - Completed Checklists
 - Missed Checklists
 - Average Score (if applicable)
 - Total Fail Points

03. Inspections & Audits Reports

3.1) Audit Score Summary

- Columns
 - Audit Template
 - Average Score
 - Highest Score
 - Lowest Score
 - Audit Count
- Charts
 - Score trend by month
 - Score distribution by location

3.2) Audit Comparison by Location

- Columns
 - Location
 - Average Score
 - Total Findings

- CAPAs Created
- Charts
 - Bar chart comparing locations
 - Radar chart (optional)

3.3) Audit Completion Status

- Columns
 - Audit Template
 - Assigned Inspector
 - Location
 - Scheduled Date
 - Completion Status
 - Not Started
 - In Progress
 - Completed
 - Pending Review
 - Overdue
 - Completion Date
 - Score
 - CAPAs Linked
- Filters
 - Date range
 - Inspector
 - Location
 - Status
- Actions
 - Send reminders
 - Reassign audit
 - Export list

3.4) Audit Findings Report

- Columns
 - Finding ID

-
- Audit Template
 - Location
 - Section
 - Question
 - Severity
 - Description
 - Assigned User
 - CAPA ID (if created)
 - Status
 - Evidence Provided
 - Charts
 - Findings by severity
 - Findings by location
 - Findings by audit section
 - Trend of findings over time
 - Filters
 - Severity
 - Location
 - Category
 - Inspector
 - Status

3.5) Non-Conformance Categories

- Columns
 - Category Name
 - Total Non-Conformances
 - Severity Breakdown
 - Linked CAPAs
- Charts
 - Category distribution pie chart
 - Category trend line chart

04. Corrective Actions (CAPA) Reports

4.1) CAPA Status Summary

- Columns:
 - Open
 - In Progress
 - Awaiting Verification
 - Closed
 - Reopened
- Pie Chart: Status distribution

4.2) CAPA Aging Report

- Columns:
 - CAPA ID
 - Severity
 - Created Date
 - Days Open
 - Assigned User
 - Overdue Status
- Actions:
 - Escalate
 - Notify assignee

4.3) CAPA Completion KPIs

- Average resolution time
- % CAPAs closed on time
- % overdue
- CAPA closure trends

05. Sensor & Temperature Reports

5.1) Temperature Compliance Report

- Columns
 - Sensor
 - Location
 - Time in Range
 - Time above threshold

- Time below threshold
 - Alerts triggered
- Charts
 - Temperature trend line
 - Compliance percentage bar chart

5.2) Sensor Alerts Summary

- Columns
 - Alert ID
 - Sensor
 - Location
 - Alert Type
 - Date/Time
 - Status
 - CAPA Linked

5.3) Sensor Performance History

- Sensor downtime
- Battery alerts
- Calibration logs

06. Label Printing Reports

- Columns
 - Template Name
 - Printed By
 - Location
 - Date & Time
 - No. of Copies
 - Batch/Expiry Values (if used)
 - Printer Used
- Filters:
 - Template
 - User
 - Location

- Date
- Actions:
 - Export print logs
 - Reprint

07. User Activity Reports

- Columns
 - User Name
 - Completed Checklists
 - Completed Audits
 - CAPAs Closed
 - Document Acknowledgements
 - Time Clock Metrics (if enabled)
- Charts:
 - Activity trend for selected users
 - Comparison between users

08. Locations Summary Report

- Metrics
 - Checklist completion rate
 - Audit score averages
 - CAPA closure rate
 - Sensor compliance
 - Document acknowledgement rate
 - Staff punctuality
- Charts:
 - Multi-location comparison bar chart
 - Heat map of high-risk locations

09. Export Center

- Export Types
 - CSV
 - Excel
 - PDF (formatted reports)

- Export Options
 - Select modules
 - Select data fields
 - Select locations
 - Select date ranges
- Recurring Export: Admins may schedule automated monthly/weekly exports.

10. Reporting Settings

- General Options
 - Default date range
 - Default comparison period
 - PDF branding (logo, header, footer)
- Access Control (Define which roles may:)
 - View all reports
 - View only module-specific reports
 - Export data
- Scheduled Reporting
 - Enable/disable
 - Frequency
 - Recipients

11. Reporting Architecture Model

The platform's reporting framework is structured into distinct analytical layers to support varying operational and decision-making needs.

- Operational Reporting Layer
 - Routine execution status
 - Failures, exceptions, missed activities
- Management Reporting Layer
 - Compliance trends and comparisons
 - Performance patterns across locations, assets, or teams
- Executive & Oversight Reporting Layer
 - Aggregate compliance posture

- Risk concentration and systemic indicators

This layered model ensures reporting consistency and prevents fragmentation into isolated dashboard views.

12. Drill-Down & Traceability Framework - All reported metrics are derived from auditable system records.

The reporting model supports logical traceability:

Metric / KPI → Operational Entity → Event / Record → Supporting Evidence

Example:

Compliance Rate → Machine / Asset → Failed Checklist → Question Response → Evidence Attachment

This structure ensures management and audit defensibility by linking aggregated indicators to verifiable operational data.

13. Trend & Comparative Analysis Capability

The reporting framework supports temporal and contextual analysis across operational datasets.

Supported analytical perspectives may include:

- Period-over-period trends
- Cross-location comparisons
- Asset / machine / team performance differentials
- Failure recurrence patterns

This enables detection of systemic issues beyond static compliance snapshots.

14. Corrective Action & CAPA Analytics

Reporting mechanisms incorporate corrective action intelligence to support compliance control and operational risk monitoring.

Key analytical dimensions may include:

- Open vs resolved CAPAs
- Closure timelines and aging
- SLA adherence and breaches
- Recurrence and root cause distributions

These metrics support management oversight in regulated and quality-sensitive industries.

15. Role-Oriented Reporting Perspectives

Reporting outputs and dashboards may be contextualized by functional roles, including:

- Operational personnel
- Supervisory roles
- Management stakeholders
- Compliance / audit functions

This ensures that reporting views align with decision responsibilities rather than presenting uniform data abstractions.

16. Report Scheduling & Distribution Logic

The platform supports automated reporting workflows to reduce operational friction and ensure timely visibility.

Reporting outputs may be:

- Generated on defined schedules
- Distributed via configured channels (e.g., email)
- Triggered by system events or thresholds

This enables proactive information delivery rather than manual retrieval dependence.

0. Template Library Module (Available when enabled)

The Template Library module provides a centralized location for **creating, managing, storing, and organizing all templates** used across the platform.

01. Template Library Dashboard

- Dashboard Widgets
 - Total Active Templates
 - Templates by Type
 - Recently Edited Templates
 - Most Frequently Used Templates
 - Templates With Pending Updates (new version available, unpublished)

- Charts
 - Template usage frequency (bar chart)
 - Templates by location assignment (pie chart)
- Actions
 - Create New Template
 - Import Template
 - Navigate to Template List

02. Template List Page

- Table Columns
 - Template Name
 - Template Type (Checklist / Audit / CAPA / Label / Custom)
 - Category
 - Assigned Locations
 - Version
 - Last Modified
 - Status (Active / Draft / Archived)
 - Actions (Edit, Duplicate, View, Publish, Archive)
- Filters & Search
 - Template type
 - Category
 - Status
 - Location
 - Created by
 - Date range
- Bulk Actions
 - Archive selected
 - Move to category
 - Export selected
 - Assign templates to locations
- Buttons
 - Create Template

- Import Template

03. Create / Edit Template (Template Builder)

3.1) Template Information Panel

- Fields:
 - Template Name
 - Template Type (Checklist, Audit, CAPA, Label, etc.)
 - Category
 - Assign to Locations (multi-select)
 - Description / Instructions
 - Tags
- Optional Settings:
 - Require Supervisor Sign-off
 - Require Photo for Fail Items (Checklist/Audit)
 - Enable Scoring (for Checklist/Audit)
 - Auto-generate CAPA on Failure (Checklist/Audit)

3.2) Section Builder (For Checklist / Audit Templates)

- Actions
 - Add Section
 - Rename Section
 - Move Section Up/Down
 - Delete Section
- Section Fields
 - Section Name
 - Section Description (optional)
 - Section Weight (for scoring templates)

3.3) Question Builder (Checklist / Audit Templates)

- Question Types
 - Yes/No
 - Pass/Fail
 - Multiple Choice
 - Checkbox

- Text Input
- Number Input
- File Upload (evidence)
- Photo Required
- Temperature Input (with min/max ranges)
- Rating Scale (1-5)
- Weighted Scoring Question
- Question Settings
 - Required / Optional
 - Show Help Text
 - Add Fail Reason Requirement
 - Attach guidance images or reference documents
 - Add scoring (points for pass/fail or numeric scoring)
 - Auto-create CAPA if failed
 - Visibility conditions (if previous question = fail → show this question)

3.4) CAPA Template Builder (Workflow Templates)

- Fields
 - CAPA Title
 - CAPA Stages: Identified, Assigned, In Progress, Completed, Verified
 - Required fields per stage: Root Cause Analysis, Corrective Action, Preventive Action, Evidence Upload, Verification Notes
 - Assign default responsible roles
 - Define escalation rules

3.5) Incident Report Template Builder

- Fields
 - Incident Title
 - Severity Levels
 - Category
 - People Involved

- Equipment Involved
- Narrative Description
- Immediate Action Taken
- Require CAPA? (toggle)

3.6) Field Validation Rules -Admins may define validation rules such as:

- Field must be numeric
- Field must be greater/less than X
- Temperature must be between X and Y
- Require minimum characters
- Mandatory vs optional fields

3.7) Preview Mode - Admins can preview:

- Mobile view (staff perspective)
- Tablet view
- Desktop/tablet inspector view
- Before publishing Actions: Save Draft, Publish Template, Save as New Version, Duplicate Template

04. Template Categories / Groups

- Category Functions
 - Create Category
 - Rename Category
 - Delete Category
 - Merge Categories
 - Assign template to category
 - Restrict category visibility by role
- Examples: Food Safety, Equipment Checks, Opening Checklists, Closing Checklists, Monthly Audits, Incident Templates

05. Template Version History

- Columns
 - Version Number
 - Date Created
 - Edited By

- Change Summary
- Status (Active / Archived)
- Actions: View Version, Restore Version, Duplicate Version
- Rules
 - Only one version can be active.
 - Restoring creates a new version.
 - Old versions remain viewable but not editable.

06. Template Permissions - Permission Matrix

- Per template, define:
 - Who can edit (Admin only / Admin + Managers)
 - Who can use (Managers / Staff / Inspectors)
 - Which locations can access the template
 - Require approvals before edits are published
- Advanced Permissions
 - Lock template (no edits allowed)
 - Enable “Draft mode only”
 - Allow only specific roles to run (execute) the template

07. Template Import / Export

- Useful for:
 - Multi-location companies
 - Standardizing best practices
 - Migrating from older systems
- Import Options
 - Import from JSON
 - Import from Excel (for checklist questions)
 - Import from supported vendor template format
- Export Options
 - Export template to JSON
 - Export printable PDF (for audit use)

08. Template Settings (Admin Only)

- General Settings

- Default location assignment
- Default scoring mode
- Enable advanced question types
- Enable conditional logic by default
- Enable template locking after publish
- Publishing Rules
 - Require manager approval before publishing
 - Auto-archive templates not used for X months
 - Auto-assign templates to new locations
- Template Usage Tracking
 - Track how many times each template is used
 - Show last execution date
 - Highlight unused templates

P. Automation & Alerts Module (Available when enabled)

Configure rule-based automations to reduce manual work (notifications, CAPA creation, escalations).

01. Automations List Page - Shows all automation rules configured by the company.

- Table Columns
 - Automation Name
 - Description
 - Trigger Type
 - Last Execution
 - Status (Active / Inactive)
 - Total Executions (count)
 - Actions (View, Edit, Disable, Duplicate, Delete, View Logs)
- Filters
 - Status
 - Trigger Type
 - Module (Checklists, Audits, CAPA, Sensors, etc.)

- Created by
- Last execution date
- Actions
 - Create New Automation
 - Duplicate Automation
 - Export Automations List

02. Create / Edit Automation (Rule Builder)

2.1) Automation Details

- Fields:
 - Automation Name
 - Description
 - Category (General / Safety / Quality / Sensor Alerts / HR / Custom)
 - Status (Active / Inactive)

2.2) Trigger Selection - Admin chooses what event starts the automation.

Trigger Categories

- Checklist Events
 - Checklist completed
 - Checklist failed
 - Question failed
 - Checklist missed
 - Checklist overdue
- Audit Events
 - Audit completed
 - Audit score below threshold
 - Non-conformance logged
- CAPA Events
 - CAPA created
 - CAPA overdue
 - CAPA moved to stage
 - CAPA closed

-
- Sensor Events
 - Temperature above threshold
 - Temperature below threshold
 - Door open too long
 - Sensor offline
 - Low battery
 - Time Clock Events
 - Late clock-in
 - Missed shift
 - Overtime triggered
 - Document Events
 - New document published
 - Document acknowledgement overdue
 - Logbook Events
 - Incident report created
 - Announcement published
 - User Activity Events
 - User inactive for X days
 - User completes training (future expansion)

2.3) Conditions Builder - Admins can define conditions to refine automation.

- Condition Examples
 - Checklist Score < 90%
 - Location = "Main Kitchen"
 - Role = "Supervisor"
 - Severity = "High"
 - CAPA Severity = "Critical"
 - Sensor Reading > 10°C for more than 5 minutes
 - User = Assigned person
 - Document type = SOP
- Condition Logic
 - ALL conditions must match (AND)

- ANY condition can match (OR)
- Nested logic groups allowed: (Condition A AND Condition B)
OR (Condition C)

2.4) Actions Builder

- **Notifications:** Send Push Notification, Send Email, Send SMS (if supported), Send In-App Alert
 - Fields:
 - Select Recipients (Users / Roles / Locations)
 - Message Title
 - Message Body (with dynamic variables)
 - Dynamic Variables Allowed:
 - Checklist Name
 - Question Failed
 - Audit Score
 - Sensor Reading
 - CAPA ID
 - Location
 - Assigned User
- **Automatic CAPA Creation**
 - Fields:
 - CAPA Template
 - Severity (Auto or fixed)
 - Assign to (User / Role)
 - Due Date Rules
 - Add notes (auto populated with event details)
- **Task / Checklist Assignment**
 - Assign a new checklist to user
 - Assign follow-up inspection
 - Create action item
 - Fields:
 - Template

-
- Location
 - Assignee
 - Deadline
 - **Sensor Rule Action**
 - Disable sensor alerts temporarily
 - Force recalibration reminder
 - Notify technician
 - **Document Actions**
 - Require acknowledgement
 - Send reminder
 - **System Actions**
 - Change CAPA stage
 - Flag entry for review
 - Generate report (for future scheduled automations)
 - **Escalation Actions** - - For overdue or ignored alerts:
 - Notify higher role
 - Escalate to supervisor
 - Escalate to admin
 - Auto-create CAPA (if not resolved)

2.5) Preview & Validation

- Before saving, system shows:
 - Trigger summary
 - Condition summary
 - Action summary
 - Warning if automation conflicts with existing ones
- Actions:
 - Save Automation
 - Activate Automation
 - Save as Draft

03. Events & Triggers Catalog - A reference page listing all events the system supports. This helps admins understand what automations they can create.

- Each item includes:
 - Trigger name
 - Module
 - Description
 - Example use cases

04. Actions Catalog - Shows every possible action that can be executed.

- Each action includes:
 - Action name
 - Description
 - Required fields
 - Supported variables

05. Automation History / Logs

- Columns
 - Automation Name
 - Trigger Event
 - Time Executed
 - Result (Success / Failed)
 - User Impacted
 - Details
 - Retry (if failed action)
- Filters:
 - Date range
 - Automation
 - Module
 - Status
- Actions:
 - View full execution trace
 - Export logs

06. Escalation Rules Page

- Escalation Types
 - CAPA escalation

- Checklist failure escalation
 - Sensor alert escalation
 - Overdue audit escalation
 - Incident escalation
- Escalation Settings
 - Notify after X minutes/hours
 - Escalate to role X
 - Escalate again after Y minutes
- Stop escalation when:
 - CAPA completed
 - Checklist completed
 - Sensor returns to normal

07. Notification Center Settings - Defines which notifications are enabled globally for the company.

- Settings
 - Enable push notifications
 - Enable email notifications
 - Enable SMS notifications
 - Allow users to customize notifications
 - Notification throttle rules (avoid spamming)
- Category Toggles - Admins can toggle notifications for:
 - Checklists
 - Audits
 - CAPA
 - Sensors
 - Scheduling
 - Documents
 - Logbook
 - Incidents

08. Automation Settings (Admin Only)

- Global Automation Rules

- Maximum automations allowed
- Default automation status (active/inactive)
- Allow managers to create automations
- Conflict detection rules
- Automation concurrency limit
- Safety Guards
 - Prevent infinite loops (Action → triggers itself)
 - Prevent duplicate notifications
 - Limit CAPA auto-creation frequency

Q. Subscription Management

01. Active Plan Overview - Displays the core subscription information:

- Plan Name
- Plan Type – Standard / Enterprise / Custom
- Start Date
- Renewal Date
- Billing Cycle – Monthly / Yearly / Custom
- Current Status – Active / Suspended / Expired / Grace Period

02. Module Entitlement

- List of all platform modules with status indicator:
 - Checklists & Forms
 - Inspections & Audits
 - Corrective Actions
 - Sensors & IoT
 - Labeling
 - Scheduling & Time Clock
 - Logbook / Communication
 - Document Management
 - Reporting & Analytics
 - Template Library
 - Automation & Alerts
- Each module displayed as:

- ON / OFF badge
- Tooltip with short description
- Non-editable by Company Admin

03. Plan Limits & Usage

- Shows entitlement versus current usage:
 - Number of Locations Included
 - Number of Active Locations Used
 - Number of Users Included
 - Number of Users Active
 - Sensors Allowed vs Connected (if applicable)
 - Automations Allowed (if applicable)
 - Storage Allocation (if applicable)
- Visual indicators:
 - Warning when nearing limit
 - Over-usage highlighted in red

04. Billing Information

- Billing Contact Name
- Billing Email
- Billing Phone (optional)
- Payment Method (masked display)
- Billing Address (if available)

05. Invoice Snapshot - Shows last 3 invoices:

- Invoice Number
- Billing Period
- Amount
- Status (Paid / Due / Overdue)
- Download PDF
- Link to external billing portal (if integrated)

06. Addons & Overages Summary - If applicable:

- Additional users
- Additional locations

- Extra sensors
- Additional storage
- Any custom add-ons

07. Actions - Primary action available to Company Admin:

- Request Plan Change
- Opens a form with:
 - Contact Name (auto-filled)
 - Email (auto-filled)
 - Current Plan
 - Requested Change
 - Upgrade Plan
 - Add Users
 - Add Locations
 - Enable Modules
 - Other
- Notes / Justification
- Submission creates:
 - Internal ticket to Super Admin
 - Notification to billing team
 - Confirmation to Company Admin

08. Functional Behaviours

- All subscription information is read-only for Company Admin.
- Module ON/OFF status reflects configuration set by Super Admin.
- Usage values must update in real time or daily sync.
- Invoice data is shown only if billing integration is enabled.
- Request Plan Change does not modify plans automatically.
- Suspended or expired plans must show a warning banner.
- Access to modules must follow the entitlement list.
- Over-usage warnings must not block current data visibility.

09. Notifications Related to Subscription - System may notify Company Admin when:

- Renewal date approaching
- Usage reaching limits
- Payment overdue
- Plan changed by Super Admin
- New module enabled/disabled

18. Staff / Manager Panel

A) Dashboard

01. Shared Features (Both Staff & Manager)

- My Tasks Summary Widget - Each count is clickable and opens My Tasks page filtered accordingly. Displays counts:
 - Assigned Checklists
 - Assigned Audits
 - CAPA Tasks
 - Document Acknowledgements
 - Follow-up Tasks
 - Today's Schedule Widget (if Time Clock enabled)
 - Shows: Shift Start / End time, Current clock status, Next scheduled shift
 - Buttons: Clock In, Clock Out
- Announcements Panel - Shows recent Logbook announcements:
 - Company messages
 - Location notices
 - Mandatory acknowledgements
- Notifications Preview - Latest alerts such as:
 - New task assigned
 - CAPA assigned
 - Document to acknowledge
 - Schedule change
 - Sensor alert
- Quick Action Buttons

- Start Checklist
- Start Audit
- My Tasks
- Clock In/Out
- Open Documents

02. Staff Only Features

- Personal Task Focus - Staff see ONLY:
 - Tasks assigned to them
 - Their own overdue items
 - Their own schedule
- Personal Performance - Optional indicators:
 - Checklists completed this week
 - On-time completion rate
 - Attendance percentage
- Restrictions - Staff CANNOT see:
 - Team tasks
 - Other users' performance
 - Approval queues
 - Location KPIs

03. Manager Only Features - Managers see all shared widgets PLUS the following:

- Team Task Overview - Displays for manager's locations:
 - Total checklists due
 - Audits pending
 - CAPAs open
 - Overdue items
 - Clickable to open Team Tasks.
- Pending Approvals - Queue of items needing manager action:
 - Audit approvals
 - CAPA verifications
 - Time clock adjustments
 - Incident reviews

- Location Performance Summary - KPIs:
 - Checklist completion rate
 - Average audit score
 - Overdue CAPAs
 - Sensor alerts count
 - Document acknowledgement %
- Staff Attendance Snapshot
 - Who is clocked in
 - Late arrivals
 - Absent staff
- Priority Alerts
 - Critical sensor breaches
 - High severity CAPAs
 - Repeated checklist failures

04. Page Layout & UI Element

- Header
 - Welcome message
 - Date & time
 - Notifications icon
 - Profile menu
- Main Layout
 - Left Column (All Users)
 - My Tasks
 - Schedule
 - Announcements
 - Right Column
 - Staff → Personal widgets only
 - Manager → Team & approval widgets
- Footer Quick Bar
 - Checklist
 - Audit

- Tasks
- Clock
- Notifications

05. Actions Available

- Staff Actions
 - Open assigned checklist
 - Open assigned audit
 - Complete CAPA
 - Acknowledge document
 - Clock In/Out
- Manager Actions - All staff actions PLUS:
 - Reassign tasks
 - Approve audits
 - Verify CAPAs
 - Approve time edits
 - View team attendance

06. Functional Behaviours

- Dashboard content must respect role permissions.
- Staff see only their own data.
- Managers see data only for assigned locations.
- Overdue items always highlighted.
- Widgets update in real time.
- Clicking widget opens filtered page.

B) My Tasks Page (Staff & Manager Panel)

01. Shared Features (Both Staff & Manager)

- Unified Task List - Displays all assigned work in one screen:
 - Checklists
 - Audits
 - CAPA actions
 - Document acknowledgements

-
- Follow-up tasks from inspections
 - Training acknowledgements (if later enabled)
 - Standard Columns
 - Task Type
 - Title / Template Name
 - Location
 - Due Date & Time
 - Priority
 - Status
 - Assigned By
 - Actions (Open / Start / View)
 - Filters
 - By Task Type
 - By Location
 - By Status (Pending / In Progress / Completed / Overdue)
 - By Due Date (Today / This Week / Custom)
 - By Priority (Low / Medium / High / Critical)
 - Search
 - Search by task name
 - Template name
 - CAPA ID
 - Location
 - Sorting
 - Due Date
 - Priority
 - Task Type
 - Location
 - Quick Actions
 - Open task
 - Mark as In Progress
 - Add quick note

- View details

02. Staff Only Features - Staff users will see ONLY their own tasks.

- Personal Task View
 - Tasks assigned directly to them
 - Tasks scheduled for their role/location
 - CAPAs assigned to them
 - Documents requiring their acknowledgement
- Overdue Section (Personal)
 - Shows only the staff member's overdue items
 - Highlighted in red
- No Access To
 - Other users' tasks
 - Reassignment options
 - Approval functions
 - Team overview

03. Manager Only Features - Managers see everything staff see PLUS team management capabilities.

- Team Tasks Tab
 - A second tab visible only to managers:
 - Tasks assigned to all users in manager's locations
 - Unassigned tasks
 - Overdue team tasks
- Reassignment Actions - Managers can:
 - Reassign a task to another user
 - Change due date
 - Adjust priority
 - Add manager notes
- Bulk Actions
 - Bulk reassign
 - Bulk change priority
 - Bulk extend due date

- Send reminders
- Review Shortcuts
 - Jump to submitted checklist for review
 - Open audit pending approval
 - Open CAPA awaiting verification

04. Page Layout & UI Elements

- Top Bar - Page title: My Tasks
 - Toggle: “My Tasks” (default for all), “Team Tasks” (manager only)
- Left Panel – Filters
 - Task Type
 - Status
 - Location
 - Date Range
 - Priority
- Main Table – Task List
- Right Panel – Task Preview (Optional)
 - Quick summary of selected task
 - Due date
 - Description
 - Start button

05. Task Types & Behaviours

- Checklist Tasks
 - Open checklist
 - Continue draft
 - Submit completion
 - Attach photos/notes
- Audit Tasks
 - Start audit
 - Record findings
 - Submit for manager approval

- CAPA Tasks
 - View CAPA details
 - Add corrective action notes
 - Upload evidence
 - Move stage (if permitted)
- Document Acknowledgement
 - Open document
 - Mark as read
 - Provide digital acknowledgement
- Follow-Ups
 - Tasks generated from failed items
 - Linked to original checklist/audit

06. Functional Behaviours & Rules

- General
 - Users must see only tasks allowed by their role.
 - Overdue tasks always appear at the top.
 - Completed tasks move to history view.
 - Tasks open in their respective execution screens.
- Staff Rules
 - Staff cannot see other users' tasks.
 - Staff cannot reassign or edit due dates.
 - Staff can only act on tasks assigned to them.
- Manager Rules
 - Managers see tasks only for their locations.
 - Managers can reassign within the same location.
 - Managers can modify priority and due dates.
- System Rules
 - Tasks sync in real time.
 - If task reassigned → notification sent.
 - If the due date changed → audit log entry created.

C) Checklist Execution

01. Shared Features (Both Staff & Manager)

- Accessing a Checklist - Users can open a checklist from:
 - My Tasks page
 - Dashboard quick action
 - Direct schedule notification
- Checklist Header Information - Displayed at top:
 - Checklist Name
 - Location
 - Date & Time
 - Assigned User
 - Instructions
 - Progress Indicator (e.g., 4/12 questions)
- Question Types Supported
 - Yes / No
 - Pass / Fail
 - Multiple Choice
 - Text Input
 - Number Input
 - Temperature Input
 - Photo Upload
 - Signature (if required)
 - Rating Scale
- Common Actions
 - Save as Draft
 - Add Notes
 - Attach Photos
 - Mark Not Applicable
 - Submit Checklist
- Failure Handling - If a question is marked as Fail:
- User may be required to:

- Enter reason
- Add photo
- Create immediate action note
- System may: Auto-create CAPA (if configured), Trigger automation alerts

02. Staff Only Features

- Personal Execution
 - Staff can only open checklists assigned to them.
 - Cannot view other users' submissions.
- Limited Editing - Cannot change:
 - Due date
 - Assigned user
 - Template content
- Submission
 - Submit for completion
 - Or submit for manager review (if required)

03. Manager Only Features - Review Mode, Managers can open completed checklists from the team.

- Actions:
 - Approve submission
 - Reject and request correction
 - Add manager comments
- Reassignment - Before completion, managers can:
 - Reassign checklist to another user
 - Change due date
 - Change priority
- Follow-Up Creation - From any failed item, manager can:
 - Create CAPA manually
 - Create follow-up task
 - Assign corrective action
- Visibility - Managers can view:

- All checklists for their locations
- Team submissions history

04. Page Layout & UI Elements

- Top Section
 - Checklist title
 - Location
 - Timer (if timed checklist)
 - Progress bar
- Main Section Questions - Each question shows:
 - Question text
 - Input control
 - Add photo button
 - Add note button
 - Fail reason field (if failed)
- Bottom Actions: Save Draft, Submit, Cancel
- For managers: Approve, Reject, Reassign

05. Functional Behaviours

- General Rules
 - Only assigned user can execute checklist.
 - Mandatory questions must be answered.
 - Photo required questions must enforce upload.
 - Failures must capture reason if configured.
 - Drafts saved automatically.
- Staff Rules
 - Staff cannot edit after submission.
 - Staff cannot approve their own checklist.
- Manager Rules
 - Managers can reopen a submitted checklist.
 - Managers can trigger CAPA from any item.
 - Managers cannot modify original answers (only comments).
- System Rules

- Auto-timestamp every answer.
- Geo-tag if enabled.
- Audit log for edits and submissions.

D) Audit Execution

01. Shared Features (Both Staff & Manager)

- Accessing an Audit
 - Audits can be opened from:
 - My Tasks page
 - Dashboard
 - Scheduled audit notification
- Audit Header Information - Displayed at top:
 - Audit Name
 - Location
 - Audit Type
 - Assigned To
 - Date & Time
 - Scoring Method (if enabled)
 - Progress Indicator
- Question & Section Types - Audits support:
 - Pass / Fail
 - Yes / No
 - Scored questions
 - Multiple choice
 - Numeric ranges
 - Photo evidence
 - Comments
 - Signature
- Evidence Capture - Users can:
 - Add photos
 - Attach files
 - Write findings

- Tag severity
- Scoring (if enabled)
 - Section scores
 - Overall score
 - Pass threshold
 - Auto-calculation
- Common Actions
 - Save as Draft
 - Add Findings
 - Mark Non-Conformance
 - Submit Audit

02. Staff or Inspector Only Features

- Conduct Audit
 - Answer all questions
 - Add evidence
 - Mark items as compliant / non-compliant
- Create Findings
 - For any failed item staff can:
 - Add description
 - Select severity
 - Attach photos
- Limitations - Staff CANNOT:
 - Approve audits
 - Modify scores after submission
 - Assign CAPA
 - View other users' audits

03. Manager Only Features

- Audit Review - Managers can:
 - Open submitted audits
 - Review answers & evidence
 - Add review comments

- Approval Workflow - Actions:
 - Approve Audit
 - Reject & Return for Correction
 - Request Additional Evidence
- CAPA Creation Findings - From any non-conformance, manager can:
 - Create CAPA
 - Assign responsible user
 - Set due date
 - Link to audit item
- Score Adjustment - If permitted:
 - Adjust weighting
 - Override score with justification
- Reassignment
 - Reassign audit before completion
 - Change inspector
 - Extend due date

04. Page Layout & UI Elements

- Top Section
 - Audit title
 - Location
 - Score summary
 - Timer (if applicable)
- Main Section - For each section:
 - Section title
 - Questions list
 - Inputs
 - Evidence upload
 - Finding panel
- Bottom Actions
 - For Staff: Save Draft, Submit
 - For Manager: Approve, Reject, Create CAPA, Reassign

05. Functional Behaviours

- General
 - Only assigned user can conduct audits.
 - Mandatory questions must be answered.
 - Evidence required fields must enforce upload.
 - Score auto-calculates based on template.
- Staff Rules
 - Staff cannot approve audits.
 - Cannot edit after submission.
 - Cannot delete findings after submit.
- Manager Rules
 - Managers can reopen audits.
 - Can create CAPA from any finding.
 - Can reassign before approval.
- System Rules
 - All actions logged.
 - Timestamps are mandatory.
 - Automations triggered on submit.

E) CAPA Task Execution

01. Shared Features (Both Staff & Manager)

- CAPA Task List Access - CAPA items appear from:
 - My Tasks page
 - Notifications
 - Direct CAPA link
- CAPA Details View - Displays:
 - CAPA ID
 - Source (Checklist / Audit / Sensor / Manual)
 - Location
 - Severity
 - Description of issue

- Due Date
- Assigned To
- Current Stage
- Common Actions - Users can:
 - Add action notes
 - Upload evidence (photos/files)
 - Update progress status
 - View history log
- Workflow Stages - Typical stages:
 - Identified
 - Assigned
 - In Progress
 - Awaiting Verification
 - Closed

02. Staff Only Features

- Execute Assigned Actions - Staff can:
 - View only CAPAs assigned to them
 - Add corrective action description
 - Upload evidence
 - Mark stage as “Completed / Ready for Verification”
- Limitations - Staff CANNOT:
 - Close CAPA
 - Change severity
 - Reassign CAPA
 - Edit original issue description

03. Manager Only Features

- CAPA Review & Verification - Managers can:
 - Review evidence
 - Accept or reject resolution
 - Add verification comments
- Reassignment - Managers may:

- Change responsible user
 - Add additional assignees
 - Modify due date
- Closure - Managers can:
 - Close CAPA
 - Reopen CAPA if insufficient
 - Escalate severity
- Link Additional Actions
 - Add follow-up task
 - Link related CAPAs
 - Attach documents

04. Page Layout & UI Elements

- Top Section
 - CAPA Title
 - ID & Source
 - Severity Badge
 - Status
 - Due Date
- Main Tabs
 - Details
 - Actions Taken
 - Evidence
 - History
- Bottom Actions
 - For Staff: Save Progress, Mark Ready for Verification
 - For Manager: Approve & Close, Reject, Reassign

05. Functional Behaviours

- General
 - Only assigned user can update CAPA.
 - Evidence required if configured.
 - All changes logged.

- Staff Rules
 - Cannot close CAPA.
 - Cannot edit source data.
- Manager Rules
 - Only managers can verify/close.
 - Rejection returns to “In Progress”.
 - Escalations trigger notifications.
- System Rules
 - Overdue CAPA triggers alerts.
 - Linked source remains referenced.

F) Logbook (Staff & Manager)

01. Shared Features (Both Staff & Manager)

- Logbook Feed - A chronological feed displaying:
 - Announcements
 - Shift notes
 - Incident reports
 - System-generated entries (e.g., CAPA created)
- Post Types Visible
 - General Note
 - Shift Handover
 - Incident Report
 - Maintenance Note
 - Safety Observation
- Common Actions
 - Read entries
 - Add comment (if allowed)
 - Acknowledge message
 - Attach photo/file
 - Filter by category/date/location
- Filters & Search
 - By type

- By location
- By date range
- By author
- Keyword search

02. Staff Only Features

- Create Entries (if permitted) Staff may:
 - Add shift notes
 - Report incidents
 - Add observations
 - Attach photos
- Acknowledgement
 - Mark announcements as read
 - Confirm receipt of instructions
- Limitations - Staff CANNOT:
 - Edit others' posts
 - Delete posts
 - Publish official announcements

03. Manager Only Features

- Create Official Announcements - Managers can:
 - Post location-wide messages
 - Set acknowledgement required
 - Pin important posts
- Moderation - Managers may:
 - Edit inappropriate entries
 - Delete posts
 - Comment with instructions
- Incident Handling - From an incident entry, managers can:
 - Create CAPA
 - Assign follow-up task
 - Escalate severity
- Visibility Control

- Choose audience (all / location / role)
- Set expiry date for posts

04. Page Layout & UI Elements

- Top Bar
 - Create Post button
 - Filters
 - Search
 - Main Feed
- Each entry shows:
 - Author
 - Date & Time
 - Type badge
 - Content
 - Attachments
 - Acknowledge button (if required)
- Side Panel
 - Pinned announcements
 - Unacknowledged items

05. Functional Behaviours

- General
 - Entries timestamped automatically.
 - Location restriction applies.
 - Attachments stored securely.
- Staff Rules
 - Can only edit own posts (if allowed).
 - Cannot delete posts.
- Manager Rules
 - Can moderate any post in their location.
 - Can require acknowledgement.
- System Rules
 - Acknowledgement tracked per user.

- Incident posts can trigger automations.

G) Document Center (Staff & Manager)

01. Shared Features (Both Staff & Manager)

- Document Library View - Users can see:
 - Document Title
 - Type (SOP / Policy / Manual / Form)
 - Version
 - Last Updated
 - Location Scope
 - Acknowledgement Required (Yes/No)
- Common Actions
 - Open document preview
 - Download file (if allowed)
 - Search documents
 - Filter by category or type
- Acknowledgement Flow - If required:
 - "I have read and understood" button
 - Date & time capture
 - Digital signature (optional)
- Filters
 - By document type
 - By category
 - By location
 - By acknowledgement status

02. Staff Only Features

- Personal Compliance View - Staff see:
 - Documents assigned to them
 - Pending acknowledgements
 - History of their acknowledgements
- Limitations - Staff CANNOT:

- Upload documents
- Replace versions

03. Manager Only Features

- Team Compliance Monitoring - Managers can view:
 - Who acknowledged
 - Who is pending
 - Overdue acknowledgements
- Reminder Actions
 - Send reminder to individual
 - Send bulk reminder
 - Escalate overdue acknowledgements
- Access to Reports
 - Download acknowledgement list
 - View by location
 - View by role

04. Page Layout & UI Elements

- Main List - Columns:
 - Document Name
 - Type
 - Version
 - Updated Date
 - My Status (Acknowledged/Pending)
 - Actions (View/Download/Acknowledge)
- Manager Additional Tab
 - “Team Status” showing: User, Document, Status, Date acknowledged

05. Functional Behaviours

- General
 - Users see only permitted documents.
 - Acknowledgement tied to version.
 - New version resets acknowledgement.

- Staff Rules
 - Staff cannot bypass acknowledgement.
 - Cannot view others' status.
- Manager Rules
 - Managers see team status only for their locations.
 - Can send reminders but not acknowledge on behalf.
- System Rules
 - All acknowledgements logged.
 - Automations may trigger on overdue.

H) Time Clock (Staff & Manager)

01. Shared Features (Both Staff & Manager)

- Clock Actions
 - Clock In
 - Start Break
 - End Break
 - Clock Out
 - Each action records: Date & Time, Location, Device (optional), Geo-tag (if enabled)
- Shift View - Users can see:
 - Today's shift schedule
 - Upcoming shifts
 - Worked hours summary
- Attendance History
 - Daily records
 - Break durations
 - Total hours
 - Status (On time / Late / Absent)
- Notifications
 - Reminder to clock in
 - Reminder to clock out

- Break limit alerts

02. Staff Only Features

- Personal Use Only - Staff can:
 - Clock in/out themselves
 - Request correction
 - View their own history
- Limitations - Staff CANNOT:
 - Edit timestamps directly
 - Approve changes
 - View other users' attendance

03. Manager Only Features

- Team Attendance View- Managers can see for their locations:
 - Who is clocked in
 - Who is late
 - Who is absent
 - Overtime status
- Adjustment Approvals - Managers may:
 - Approve edit requests
 - Add missed punch
 - Correct break time
- Attendance Actions
 - Force clock out (if permitted)
 - Add manual entry with reason
 - Export attendance
- Alerts
 - Late arrivals
 - No-show alerts
 - Overtime warnings

04. Page Layout & UI Elements

- Top Section
 - Current status badge

- Clock In/Out button
- Shift info
- Main Tabs
 - Today
 - Schedule
 - History
 - Requests (manager)
- Manager Panel
 - Team list with status
 - Approvals queue

05. Functional Behaviours

- General
 - One active session per user.
 - Breaks must be within shift.
 - Geo restrictions if enabled.
- Staff Rules
 - Cannot modify records.
 - Must request corrections.
- Manager Rules
 - Only managers approve edits.
 - Can manage only their locations.
- System Rules
 - All actions logged.
 - Automations may trigger on late.

I) Notification Center

01. Shared Features (Both Staff & Manager)

- Notification List - Each notification shows:
 - Title
 - Message
 - Source Module

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- Date & Time
 - Priority
 - Status (Unread/Read)
 - Actions - Users can:
 - Open related item (checklist, CAPA, audit, document)
 - Mark as read
 - Delete notification (if allowed)
 - Filter notifications
 - Categories - Notifications grouped by:
 - Tasks
 - CAPA
 - Audits
 - Sensors
 - Documents
 - Schedule
 - System messages
 - Filters & Search
 - By category
 - By date
 - By priority
 - Keyword search
 - Delivery Channels - Depending on system settings:
 - In-app
 - Push
 - Email

02. Staff Only Features

- Personal Alerts Only - Staff receive:
 - Assigned task alerts
 - CAPA assignments
 - Document acknowledgement requests
 - Shift reminders

- Sensor alerts relevant to them
- Limitations - Staff CANNOT:
 - View other users' notifications
 - See team-level escalations
 - Modify notification rules

03. Manager Only Features

- Team Alerts - Managers receive:
 - Team overdue tasks
 - Audit submissions pending approval
 - CAPA awaiting verification
 - Staff late clock-ins
 - Critical sensor alerts
- Escalation View
 - View escalated alerts
 - Reassign responsibility
 - Trigger follow-up actions
- Bulk Actions
 - Mark team notifications as handled
 - Send reminders to users

04. Page Layout & UI Elements

- Top Bar
 - Unread count
 - Filter dropdown
 - Search
- Main List
 - Notification cards with icons by module
 - Priority color codes
- Right Panel (optional)
 - Preview of selected notification
 - Quick action buttons

05. Functional Behaviours

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- General
 - Notifications linked to source item.
 - Clicking opens correct page.
 - Read status synced across devices.
 - Staff Rules
 - Only personal notifications shown.
 - Cannot access team alerts.
 - Manager Rules
 - Managers see team alerts for their locations.
 - Can act on escalations.
 - System Rules
 - Notifications follow automation rules.
 - Throttling prevents spam.

J) Profile Settings

01. Shared Features (Staff & Manager)

- Profile Information - Users can view/edit:
 - Full Name
 - Phone Number
 - Email (if allowed)
 - Profile Photo
 - Preferred Language
 - Time Zone (if enabled)
- Security
 - Change Password
 - Enable MFA (if supported)
 - View last login activity
 - Active sessions list
- Notification Preferences
 - User can control: In-app notifications, Email alerts, Push alerts, SMS (if enabled)

- Per category: Tasks, CAPA, Audits, Sensors, Documents, Schedule
- Display Preferences
 - Dark / Light mode
 - Font size
 - Default landing page
- Help & Support
 - User guide link
 - Contact support
 - App version info

02. Staff Only Features - Profile functions are identical for both roles.

- Staff cannot:
 - Change role
 - Change permission
 - Access other users

03. Manager Only Features - managers manage only their personal profile here.

Team settings are handled in Company Admin panel, not here.

04. Page Layout & UI Elements

- Tabs
 - Profile
 - Security
 - Notifications
 - Preferences
 - Activity Log
- Actions
 - Save Changes
 - Change Password
 - Logout All Sessions

05. Functional Behaviours

- Users can edit only their own profile.
- Email change may require verification.

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- Password must follow policy.
 - Notification changes apply immediately.
 - Activity log read-only.